

Mark schemes

Q1.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Calculates the correct sample mean	AO1.1b	B1	$\bar{x} = \frac{3540}{100}$ $= 35.4$
	Finds correct z value to at least 3 significant figures. Condone -2.326... Can be implied by a correct confidence interval.	AO1.1a	M1	$z = 2.32634757$ $\bar{x} \pm z \sqrt{\frac{\sigma^2}{n}}$
	Uses formula for confidence interval with $\sqrt{\frac{10}{100}}$	AO1.1a	M1	$= 35.4 \pm 2.326 \sqrt{\frac{10}{100}}$ $= (34.66, 36.14) \text{ or } 35.4 \pm 0.7357$
	Obtains confidence interval CAO	AO1.1b	A1	
(b)	States null hypothesis is rejected as the confidence interval does not contain 38. OE Follow through their confidence interval.	AO3.5a	E1F	Dante rejects the null hypothesis because 38 is outside the confidence interval.
				Total 5 marks

Q2.

	Marking Instructions	AO	Marks	Typical Solution
(a)	States 1.83 AWRT	AO1.1b	B1	1.83
(b)	Comments that David's statement is not or may be not true	AO2.3	R1	David's statement may not be true because
	Mentions that interval is based on mean from sample or Mentions that interval is for population mean length, not for population of lizard lengths	AO2.4	R1	Interval is centred on sample mean, not population mean or interval is for population mean length, not lizard lengths
(c)	Uses correct expression in b , s and n in equation	AO1.1a	M1	$\frac{b \times s}{\sqrt{n}} = 4.27$ or $2 \times$

relating to width value			$\frac{b \times s}{\sqrt{n}} = 8.54$
Solves for b FT incorrect s	AO1.1b	A1F	$\frac{b \times 4.15(45691)}{\sqrt{10}} = 4.27$ $b = 3.25$
Deduces that $\beta = 99$ (condone 99 %)	AO2.2a	R1	From t tables, $\beta = 99$
Total 6 marks			

Q3.

- (a) Sample mean = 53.06, $s = 1.140$

Both. For s AWRT 1.14

B1

$t_5 = 2.571$

AWRT 2.57

B1

Sample mean $\pm t \times s / \sqrt{6}$

For $\sqrt{6}$

M1

Rest of formula. Allow $t_5 = 2.01$ to 2.02 , or $t_6 = 2.45$

m1

$(53.06 \pm 1.20) = (51.86, 54.26)$

Either form ± 0.01 in total.

A1

5

- (b) Sample mean is lower than last year's mean so claim **may** be true.
 53.41 lies within c.i. so **not certain** that mean time is better.
 Performance in competition does not depend on mean time.
 Times seem to be improving.

*E1 each for sensible comments either supporting or against statement up to a maximum of 2.
 Comment must be **uncertain**.*

E2

2

[7]

Q4.

B1

CI for μ is $\bar{x} \pm z \times \frac{\sigma}{\sqrt{n}}$
 Used with z (2.05 to 2.58),
 $\bar{x}$ (19.9) and σ (0.4)
 and $\div \sqrt{n}$ with $n > 1$

M1

Thus $19.9 \pm 2.3263 \times \frac{0.4}{\sqrt{25}}$
 z (2.05 to 2.06 or 2.32 to 2.33 or 2.57 to 2.58),
 $\bar{x}$ (19.9) and σ (0.4) and $\div \sqrt{25 \text{ or } 24}$

A1

Hence $\underline{19.9 \pm 0.2}$
 $CAO / AWRT$ (0.186104)

or

$\underline{(19.7, 20.1)}$
 $AWRT$

A1

4

- (b) **Clear correct comparison of 20 with CI**
F on CI providing it contains 20

eg 20 is within CI **or** $LCL < 20 < UCL$
Quoting values for CI is **not** required

BF1

SO

Agree with claim **or** no reason to doubt claim
OE; dependent on previous BF1

Bdep1

2

[6]

Q5.

(a) (i) $\bar{x} = \frac{497.5}{25} = \underline{\underline{19.9}}$

CAO

B1

$$98\% (0.98) \Rightarrow z = \underline{2.32 \text{ to } 2.33}$$

AWFW (2.3263)

B1

CI for μ is $\bar{x} \pm z \times \frac{\sigma}{\sqrt{n}}$

Used with z (2.05 to 2.58),
 $\bar{x}$ (497.5 or 19 to 21) and σ (0.4)
 and $\div \sqrt{n}$ with $n > 1$

M1

Thus $19.9 \pm 2.3263 \times \frac{0.4}{\sqrt{25}}$

z (2.05 to 2.06 or 2.32 to 2.33 or 2.57 to 2.58),
 $\bar{x}$ (19.9) and σ (0.4) and $\div \sqrt{25 \text{ or } 24}$

A1

Hence $\underline{19.9 \pm 0.2}$

CAO / AWRT (0.186104)

or

$\underline{(19.7, 20.1)}$

AWRT

A1

(ii) **Clear correct comparison of 20 with CI**

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 F on CI providing it contains 20

eg 20 is within CI **or** $LCL < 20 < UCL$

Quoting values for CI is **not** required

BF1

Agree with claim **or** no reason to doubt claim

OE; dependent on previous BF1

Bdep1

2

(b) (i) $Y \sim N(25.25, 0.35^2)$

Accept percentage equivalent probabilities

$V(\text{mean}) = \underline{0.35^2 / 10 \text{ or } 0.0122 \text{ to } 0.0123}$

CAO / AWFW (0.01225)

or



$$\text{SD (mean)} = \frac{0.35}{\sqrt{10}} \text{ or } 0.11 \text{ to } 0.111$$

CAO / AFWW (0.11068)

B1

$$P(\bar{Y} < 25) = P\left(Z < \frac{25 - 25.25}{0.35/\sqrt{10}}\right)$$

Standardising 25 using 25.25 and $0.35/\sqrt{10}$ OE
but allow (25.25 - 25)

M1

$$= P(Z < -2.25877) = 1 - P(Z < 2.25877)$$

Correct area change

May be implied by a **correct** answer **or** an answer < 0.5

m1

$$= 1 - (0.98809 \text{ to } 0.98778)$$

$$= \underline{0.011 \text{ to } 0.013}$$

AWFW (0.01195)

(0.987 to 0.989) $\Rightarrow$ B1 M1 m0 A0

A1

4

(ii)

$$P(Y > 25) = P\left(Z > \frac{25 - 25.25}{0.35}\right)$$

Standardising 25 using 25.25 and 0.35 but allow (25.25 - 25)

M1

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$$= P(Z > -0.71429) = P(Z < 0.71429)$$

$$= \underline{0.761 \text{ to } 0.764}$$

AWFW (0.76247)

(0.236 to 0.239) $\Rightarrow$ M1 A0

A1

$$P(Y > 25 \text{ in each of } 10) = p^{10}$$

Any p^{10} providing $0 < p < 1$

May be implied by a **correct** answer

M1

$$= \underline{0.065 \text{ to } 0.068}$$

AWFW (0.06641)

A1

4

Q6.

(a) $\bar{x} = 948$ and $s^2 = 4817.25$

Both; AWR T 4820 ($s = 69.406$)

B1

$t_8 = 2.896$

AWRT 2.90

B1

C.I. = $948 \pm 2.896 \times \sqrt{\frac{4817.25}{9}}$

For division by $\sqrt{9}$

M1

For rest of expression, must be t_8 or t_9 ($= 2.821$)

m1

$= 948 \pm 67.0 = (881, 1015)$

Either form AWR T ± 67

Accept 1010 or 1020 as upper limit

A1

5

(b) (i) $(927 + 1063) \div 2 = 995$

CAO

B1

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1

(ii) **Dependent on partial overlap**

Because of the overlap by the confidence intervals ...

E1

... no definite conclusion is possible

Accept "No evidence"

Edep1

2

SC

Reference to evidence provided by the mean or the limits being lower 'suggesting' or 'providing evidence' or 'supporting' weight reduction scores 1

The statement must be not definite. Anything definite, eg. 'proves that' or 'shows that' scores 0

(E1)

Q7.

(a) $\bar{x} - ns = (45.8 - n \times 24.0) < 0$

SC: Accept quoted values of $(-4 \text{ to } -1)$ ($n = 2$) or $(-28.5 \text{ to } -23.5)$ ($n = 3$) (both AWWF) and

Allow (45 to 47) and any multiple of (23.5 to 24.5) which gives value < 0

Must clearly state the value of a numerical expression

M1

negative salaries are impossible

OE; must be in context

Negative values impossible $\Rightarrow A0$

A1

2

Alternative solution 1

$P(X < 0 \mid N(45.8, 24.0^2)) = P(Z < -1.91)$

Standardising 0 using 45.8 & 24.0

(M1)

$= 0.027 \text{ to } 0.03$

In addition to probability within range, must state that negative salaries are impossible

(A1)

(2)

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Alternative solution 2

$P(X > 60 \mid N(45.8, 24.0^2)) = P(Z > 0.59)$

Standardising 60 using 45.8 & 24.0

(M1)

$= 0.27 \text{ to } 0.28$

In addition to probability within range, must compare calculated value to $6/50 = 0.12$ OE

(A1)

(2)

- (b) (i) Large sample or $n > 25$ or 30 or $n = 50$
so

OE

B1

CLT applies

Must indicate CLT; dependent on B1

Indication that other than sample mean is normally distributed $\Rightarrow$ Bdep0

Bdep1

2

(ii) 99% (0.99) $\Rightarrow z = 2.57$ to 2.58

AWFW (2.5758)

B1

CI for μ is $\bar{x} \pm z \times \frac{s}{\sqrt{n}}$
Used with (45.8 & 24.0) and
 $z(1.64$ to $2.58)$ & $\div \sqrt{n}$ with $n > 1$

M1

Thus $45.8 \pm 2.5758 \times \frac{24.0}{\sqrt{50}}$
F on $z(1.64$ to 1.65 or 2.32 to 2.33
or 2.57 to $2.58)$ and $\div \sqrt{50}$ or 49

AF1

Hence $45.8 \pm (8.7$ to $8.8)$ or $45800 \pm (8700$ to $8800)$

OR

(37.0) to 37.1 , 54.5 to $54.6)$ or $(37000$ to 37100 , 54500 to $54600)$

CAO / AFWW (8.74)

Ignore (absence of) quoted units

AWFW

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A1

4

(c) Clear correct comparison of 55 or 55000 with c's UCL or CI

Accept 55000 compared with c's 54.5 to 54.6

(ie different units)

B1

$(6/50$ or 0.12 or $12\%) </\neq 0.25$ or 25%

OE; correct comparison mentioning both 12% and 25%

B1

Reject both / each of the two claims

Dependent on B1 B1

Bdep1

3

Additional comment illustrations

It / (claimed) mean / (claimed) value > UCL / CI
Must indicate 55 or 55000

(B0)

99% have (mean) weights between CLs so...

(B0)

Any comparison of 60 (£60 000) with UCL / CI
Value of 60 does not refer to mean

(B0)

$P(\bar{X} > 60 \mid N(45.8, 24.0^2)) = P(Z < 0.59)$
 $= (0.27 \text{ to } 0.28) > 6/50 = 0.12$
Assumes salaries ~ N; cf (a)(ii)

(B0)

[11]

Q8.

(a) (i) $\bar{x} = \frac{2290}{50} = 45.8 \text{ or } 45800$

CAO

B1

$(s^2 =) \frac{28225.5}{49 \text{ or } 50} \text{ or } (s =) \sqrt{\frac{28225.5}{49 \text{ or } 50}}$

Ignore notation

M1

$s = 24(.0) \text{ or } 24000 \text{ to } 24001$
AWRT / AFWW (24.00064)
($\sigma = 23.75942$)

SCs: (for no seen working)
M1 A1 for 24.0 or 24000 to 24001
M1 A0 for 24 or 23700 to 23800

A1

3

(ii) $\bar{x} - ns = (45.8 - n \times 24.0) < 0$
SC: Accept quoted values of
 $(-4 \text{ to } -1) (n = 2) \text{ or } (-28.5 \text{ to } -23.5) (n = 3)$ (both AFWW)
Allow (45 to 47) and any multiple of (23.5 to 24.5) which gives value < 0

Must clearly state the value of a numerical expression

M1

and negative salaries are impossible

OE; must be in context

Negative values impossible $\Rightarrow$ A0

A1

2

Alternative solution 1

$$P(X < 0 \mid N(45.8, 24.0^2)) = P(Z < -1.91)$$

Standardising 0 using 45.8 & 24.0

(M1)

$$= 0.027 \text{ to } 0.03$$

*In addition to probability within range,
must state that negative salaries are impossible*

(A1)

(2)

Alternative solution 2

$$P(X > 60 \mid N(45.8, 24.0^2)) = P(Z > 0.59)$$

Standardising 60 using 45.8 & 24.0

(M1)

$$= 0.27 \text{ to } 0.28$$

*In addition to probability within range,
must compare calculated value to $6 / 50 = 0.12$ OE*

(A1)

(2)

(b) $99\% (0.99) \Rightarrow z = 2.57 \text{ to } 2.58$

AWFW (2.5758)

B1

CI for μ is $\bar{x} \pm z \times \frac{s}{\sqrt{n}}$

*Used with ($\bar{x}$ & s) from (a)(i) and
 $z(1.64 \text{ to } 2.58)$ & $\div \sqrt{n}$ with $n > 1$*

M1

Thus $45.8 \pm 2.5758 \times \frac{24.0}{\sqrt{50}}$
*F on ($\bar{x}$ & s) with $\div \sqrt{50 \text{ or } 49}$ &
 $z(1.64 \text{ to } 1.65 \text{ or } 2.32 \text{ to } 2.33 \text{ or } 2.57 \text{ to } 2.58)$*

AF1

Hence $45.8 \pm (8.7 \text{ to } 8.8)$
or $45800 \pm (8700 \text{ to } 8800)$
OR
 $(37.0) \text{ to } 37.1, 54.5 \text{ to } 54.6$
or $(37000 \text{ to } 37100, 54500 \text{ to } 54600)$

CAO / AFWW (8.74)
Ignore (absence of) quoted units
AWFW

A1

4

(c) See supplementary sheet for additional illustrations

Clear correct comparison of 55 or 55000 with c's UCL or CI
Accept 55000 compared with
c's 54.5 to 54.6 (ie different units)

B1

$(6/50 \text{ or } 0.12 \text{ or } 12\%) < \neq 0.25 \text{ or } 25\%$
OE; correct comparison mentioning both 12% and 25%

B1

Reject both / each of the two claims
Dependent on B1 B1

Bdep1

3

Additional comment illustrations

It / (claimed) mean / (claimed) value > UCL / CI
Must indicate 55 or 55000

(B0)

99% have (mean) weights between CLs so...

(B0)

Any comparison of 60 (£60 000) with UCL / IC
Value of 60 does not refer to mean

(B0)

$P(X < 60 \mid N(45.8, 24.0^2)) = P(Z > 0.59)$
 $= (0.27 \text{ to } 0.28) < 6 / 50 = 0.12$
Assumes salaries ~ N; cf (a)(ii)

(B0)

[12]

Q9.

- (a) 98% (0.98) $\Rightarrow z = \frac{2.32 \text{ to } 2.33}{\text{AWFW} \quad (2.3263)}$

B1

CI for μ is $\bar{x} \pm z \times \frac{\sigma}{\sqrt{n}}$

Used with z (2.05 to 2.58), $\bar{x}$ (5.05, 5050 or 181.8),
 σ (0.0075, 0.075, 0.75, 7.5 or 75) and $\div \sqrt{n}$ with $n > 1$

M1

Thus $5.05 \pm 2.3263 \times \frac{0.075}{\sqrt{36}}$

z (2.05 to 2.06 or 2.32 to 2.33 or 2.57 to 2.58),
 $\bar{x}$ (5.05) & σ (0.075) or x (5050) & σ (75)
and $\div \sqrt{36 \text{ or } 35}$

A1

Hence 5.05 $\pm$ 0.03 or 5050 $\pm$ 30
OR
(5.02, 5.08) or (5020, 5080)

CAO / AWR T

Dependent on previous A1 so can be
scored with $z \neq 2.32 \text{ to } 2.33$

Ignore (absence of) quoted units

AWRT to 3sf accuracy

Adep 1

4

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Note Use of t (2.43 to 2.72) $\Rightarrow$ B1 B0 M1 A0 A0 max

- (b) **Clear correct comparison of 5 or 5000 with LCL or CI**

SO

agree with (first) claim (about mean)

Dependent on Adep1 in (a)

Must use consistent units

Adep 1

(8/36 or 0.22 or 22%) v (1/10 or 0.10 or 10%)

or 8 v 3.6 (3 to 4)

Mention of a value on LHS **and** a value on RHS

B1

SO

8/36 OE > / $\neq$ 1/10 OE

SO

disagree with (second) claim (about individuals)

Dependent on B1

Explicit comparison of values and correct conclusion

Bdep 1

3

Notes 1 It / (claimed) mean / (claimed) value < LCL / CI $\Rightarrow$ Adep0
Must indicate 5 or 5000

2 98% have (mean) weights between CLs so ... $\Rightarrow$ Adep0

3 Any reference to CI for second claim $\Rightarrow$ B0 Bdep0 Claim refers to individual bottles

[7]

Q10.

(a) $\bar{x} = \frac{\sum x}{n} = \frac{546}{15} = \frac{182}{5} = 36.4$

oe

B1

$$s^2 = \frac{\sum (x - \bar{x})^2}{n-1} = \frac{1407.6}{14} = 100.54 \text{ (or } s = 10.03)$$

$$\sigma^2 = 93.84 \text{ or } \sigma = 9.687$$

if $\frac{\sigma}{\sqrt{14}}$ used below

B1

$$t_{crit} = \pm 2.624$$

ignore signs for t_{crit} (allow $t = 2.62$)

(if z used then **max** (B1B1B0 M0A0A0))

B1

$$98\% \text{ CI for } \mu : 36.4 \pm 2.624 \times \frac{s}{\sqrt{15}}$$

$$\text{their } \bar{x} \pm t_{14} \times \frac{\text{their } s}{\sqrt{15}} \text{ or their } \bar{x} \pm t_{14} \times \frac{\text{their } \sigma}{\sqrt{14}}$$

M1

$$\left. \begin{array}{l} (29.6, 43.2) \\ 36.4 \pm 6.8 \end{array} \right\}$$

(allow any of the following for t_{14} :

1.345; 1.761; 2.145; 2.624; 2.977)

A1ft

$$= 29.6, 43.2$$

cao

A1

6

(b) $40.0 \in \text{C.I.}$

Must refer to 40 (dep M1)

E1ft

$\Rightarrow$ no change;

Dep on previous mark

E1ft

2

[8]

Q11.

(a) (i) Mean, $\bar{x} = 20.4$
CAO

B1

1

(ii) Standard deviation $s = \sqrt{\frac{400.24}{64}}$
Expression must be seen $(\sqrt{6.25375})$

M1

$= 2.50$

AWRT (2.50075)

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A1

2

NB: $s = \frac{400.24}{64 \text{ or } 65}$ (6.15754 or 6.25375)

No $\sqrt{\quad}$ and / or use of divisor n

(B1)

or

$s = \frac{\sqrt{400.24}}{65}$ (2.48144)

Use of divisor n

(B1)

(b) $96\%(0.96) \Rightarrow z = 2.05 \text{ to } 2.06$

AWFW (2.0537)

B1

CI for μ is $\bar{x} \pm z \times \frac{s}{\sqrt{n}}$

Used

Must have $\sqrt{n}$ with $n > 1$

M1

Thus $20 \pm 2.0537 \times \frac{2.50}{\sqrt{65 \text{ or } 64}}$
F on $\bar{x}$ and z

A1F

Hence 20.4 ± 0.6 or (19.8, 21(.0))
AWRT

A1

4

- (c) CI in (b)(i) contains/includes 20
Or equivalent
Dependent on CI in (b)(i)

B1F

thus

no (significant) evidence to support claim
Or equivalent
Dependent on B1F

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Bdep1

2

[9]

Q12.

(a) (i) $\frac{0.98 + 1.00}{2}$ or $\frac{0.975 + 1.005}{2}$ or
 $0.98 + \frac{0.02}{2}$ or $0.975 + \frac{0.03}{2} = 0.99$

AG

(At least) one correct expression seen

Ignore contradictions

Accept any valid equivalent

B1

(ii) $\frac{0.97 + 0.98}{2} = 0.975$ and $\frac{1.00 + 1.01}{2} = 1.005$

Both CAO
Can not be implied from (a)(i)

B1

2

SC In (a)(i) and (a)(ii) allow $1.004\dot{9}$ or 1.0049... etc
Similar forms for lower boundary

(b) Mean, $\bar{x} = 1.062$

CAO $\sum fx = 106.2$ Ignore notation

B1

Standard deviation, s or $\sigma = 0.043$

AWRT $\sum fx^2 = 112.9662$

If B0 B0, M1 can be awarded for attempt at $\frac{\sum fx}{100}$

B2

3

(c) (i) $99\%(0.99) \Rightarrow z = 2.57$ to 2.58
AWFW (2.5758)

B1

$t_{99}(0.995) = 2.626$ AWRT

(B1)

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CI for μ is $\bar{x} \pm (z \text{ or } t) \times \frac{(s \text{ or } \sigma)}{\sqrt{n}}$

Used

Must have $\sqrt{n}$ with $n > 1$

M1

Thus $1.062 \pm 2.5758 \times \frac{0.043}{\sqrt{100 \text{ or } 99}}$

F on $\bar{x}$, s/σ and z/t

A1F

Hence 1.06 ± 0.01 or $(1.05, 1.07)$

AWRT; award even if previous inaccuracies in $\bar{x}$,
 s/σ or z/t

A1

4

- (ii) Sample data grouped
 σ unknown

Exact sample values unknown / midpoints used
 s calculated from a sample

$\bar{x}$ and s calculated from grouped data
 $\bar{x}$ (not μ) and s are estimates
NOT data values rounded

B1

1

- (d) (i) CI for μ or CI in (c)(i) > 1
LCL of CI for μ or
LCL of CI in (c)(i) > 1
Or equivalent; must compare CI to 1
Dependent on CI in (c)(i) > 1

B1

- (ii) 99 or 100 or all sample/ table/ data volumes/ values/
x-values/ cartons are within this range
(or none / 0 or 1 volumes outside)

B1

2

[13]

Q13.

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(a)

$$\left. \begin{array}{l} Y \sim N(\mu, 4) \\ n = 16, \bar{y} = 450 \end{array} \right\}$$

(known variance) use $\Rightarrow z$

For 95% CI $z_{crit} = 1.96$

B1

$$\left. \begin{array}{l} 450 \pm 1.96 \times \frac{2}{\sqrt{16}} \\ 450 \pm 0.98 \end{array} \right\}$$

$$450 \pm 0.98$$

M1

(449, 451)

awrt

A1

3

(b) (i) $X \sim N(\mu_x, \sigma^2)$

(unknown variance) $\Rightarrow$ use t_{n-1}

$$\left. \begin{aligned} n=9 \quad \& \quad \bar{x} = \frac{4950}{9} = 550 \\ s^2_{n-1} = \frac{334}{8} = 41.75 \quad (s_{n-1} = 6.461) \end{aligned} \right\}$$

both

B1

For 90% CI $t_{crit} = 1.860$

$$550 \pm 1.860 \times \frac{\sqrt{41.75}}{\sqrt{9}}$$

$\left\{ \begin{array}{l} \text{their } \bar{x} \pm t_8 \times \frac{\text{their } s_{n-1}}{\sqrt{9}} \\ \text{(must have at } t_8 \text{ - value)} \end{array} \right.$

B1

M1

$$550 \pm 4.0$$

A1 ft

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(546, 554)

awrt

A1

5

(ii) 545 not in 90% CI

$\therefore$ Reject claim

Evidence to suggest that **mean** content of a bottle of tomato sauce is not 545 grams and hence to reject Holly's claim

Alternative

$$H_0 : \mu_x = 545$$

$$H_1 : \mu_x \neq 545$$

$$t = \frac{550 - 545}{\sqrt{\frac{41.75}{9}}} = 2.32$$

B1ft
(dep on (b)(i))

at 10% level of significance.

$$\left. \begin{array}{l} t_{\text{crit}} = 1.86 < 2.32 \\ \text{Reject } H_0 \end{array} \right\} \quad \text{B1}$$

Comment in context **E1**

E1 ft
(dep on (b)(i))

10% level of significance.

B1

3

Alternatives (such as):

Claim justified at 1% level of significance

$$t_{\text{crit}} = 3.355 > 2.32$$

Accept H_0
1% significance level

[11]

Q14.

(a) (i) 96% (0.96) $\Rightarrow z = 2.05$ to 2.06
AWFW (2.0537)

B1

$$\bar{x} \pm z \times \frac{s}{\sqrt{n}}$$

CI for μ is
Used with 251.1 and 1.94 correctly
Must have $\sqrt{n}$ with $n > 1$

M1

Thus

$$251.1 \pm 2.0537 \times \frac{1.94}{\sqrt{50 \text{ or } 49}}$$

F on z only

AF1

Hence 251.1 ± 0.6
or $(250.5, 251.7)$

CAO/AWRT

Dependent on AF1 but not on z so can be gained using an incorrect z

AWRT

Adep1

4

- (ii) Claim is $\mu > 250$

Clear correct comparison of 250 with LCL or CI

F on CI (250 < LCL or CI)

BF1

so

Claim is supported/reasonable/correct/true/etc

Must be consistent with c's comparison

Dependent on BF1

Bdep1

2

- (b) (i) I for x is $\bar{x} \pm z \times s$
 $= 251.1 \pm 2.0537 \times 1.94$

Must have $n = 1$ and correct or same z as in (a)(i)

M1

Hence

$$251.1 \pm 4(.0)$$

CAO/AWRT

or

$$(247, 255)$$

*F on z in (a)(i); can be gained using an incorrect z
 AWRT*

AF1

2

- (ii) Some individual packets are likely to/will contain less than 250 grams

Or equivalent F on (b)(i)

BF1

1

[9]

Q15.

(a) $\sqrt{\frac{184.5}{49}}$ or $1.92 \times \sqrt{\frac{50}{49}}$

Fully correct expression or equivalent must be seen

Note: $s = \sqrt{184.5/50} = 1.939 \Rightarrow B0$

B1

$$= 1.94$$

AG

1

- (b) (i) 96% (0.96) $\Rightarrow z = 2.05$ to 2.06

AWFW

B1

$$\bar{x} \pm z \times \frac{s}{\sqrt{n}}$$
 CI for μ is
 Used with 251.1 and 1.94 correctly
 Must have $\sqrt{n}$ with $n > 1$

M1

Thus
$$251.1 \pm 2.0537 \times \frac{1.94}{\sqrt{50 \text{ or } 49}}$$

F on z only

AF1

Hence 251.1 ± 0.6
 or $(250.5, 251.7)$
 CAO/AWRT
 Dependent on AF1 but not on z so can be gained using an incorrect z
 AWRT

Adep1

4

(ii) Claim is $\mu > 250$

Clear correct comparison of 250 with LCL or CI
F on CI (250 < LCL or CI)

BF1

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so
 Claim is supported/reasonable/correct/true/etc
 Must be consistent with c's comparison
 Dependent on BF1

Bdep1

2

(c) $\bar{x} - ns = 251.1 - n \times 1.94 < 250$
SC: Quoted values of 249.2, 247.2 or 245.3
 (AWRT) $\Rightarrow$ M1
 Allow any multiple of 1.94
 Must **clearly indicate** the value of a numerical expression giving a result less than 250

M1

so
 Some individual packets are likely to/will contain less than 250 grams

Or equivalent

A1

2

[9]

Q16.

- (a) Represent a random sample

Or equivalent

B1

$$\text{Mean} = \bar{x} = \frac{3155}{50} = 63.1$$

CAO

B1

$$\text{Variance} = s^2 = \frac{7180.5}{49} = 146.5 \text{ to } 146.6$$

$$\text{AWFW} \left(\frac{7180.5}{50} = 143.61 \right)$$

B1

3

- (b) 99% (0.99) $\Rightarrow z = 2.57$ to 2.58

AWFW (2.5758)

$t_{49} (0.995) = 2.67$ to 2.68

B1

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$$\text{CI for } \mu \text{ is } \bar{x} \pm z/t \times \frac{s}{\sqrt{n}}$$

Used

Must have $\sqrt{n}$ with $n > 1$

M1

$$\text{Thus } 63.1 \pm 2.5758 \times \frac{\sqrt{146.5}}{\sqrt{50}}$$

F on $\bar{x}$, s^2 and z/t

A1F

Hence $63.1 \pm (4.39 \text{ to } 4.42)$

or (58.6 to 58.8, 67.4 to 67.6)

CAO & AFWW

Dependant on A1F with correct s^2 / s

AWFW (58.7, 67.5)

A1dep

Note:

Use of t gives $63.1 \pm (4.57 \text{ to } 4.59)$
or $(58.5 \text{ to } 58.6, 67.6 \text{ to } 67.7)$

4

- (c) 60 minutes is within C1
Or equivalent; F on (b)
Any reference to 63.1 $\Rightarrow$ B0F

B1F

Wyn's belief doubtful/incorrect

or

Wyn's belief may be correct

Or equivalent

Dependent on B1F

B1 dep

2

- (d) 1/100 or 0.01 or 1%
CAO; not 0.01%

B1

1

[10]

Q17.

- (a) 98% (0.98) $\Rightarrow z = 2.32 \text{ to } 2.33$
AWFW (2.3263)

B1

CI for μ is $\bar{x} \pm z \times \frac{\sigma}{\sqrt{n}}$

Used

Must have $\sqrt{n}$ with $n > 1$

M1

Thus $1030 \pm 2.3263 \times \frac{50}{\sqrt{20}}$
F on z only

A1F

Hence 1030 ± 26

CAO & AWRT

or (1004, 1056)

AWRT

A1

4

- (b) Whole of confidence interval is **above 1000**

F on (a)

Or equivalent

B1F

so

Agree with claim

F on (a)

Or equivalent

Dependent on previous B1F

B1Fdep

2

[6]

Q18.

- (a) 12 elephants are a **random sample**
OR are **selected independently**

OE; eg representative

B1

$$\text{Mean } \bar{x} = \frac{39.24}{12} = 3.27$$

CAO

B1

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$$98\% \Rightarrow z = 2.32 \text{ to } 2.33$$

AWFW (2.3263)

B1

$$\text{CI for } \mu \text{ is } \bar{x} \pm z \times \frac{\sigma}{\sqrt{n}}$$

Used; must have $\sqrt{n}$ with $n > 1$

M1

$$\text{Thus } 3.27 \pm 2.3263 \times \frac{0.20}{\sqrt{12}}$$

ft on $\bar{x}$ and z only

A1ft

Hence 3.27 ± 0.1343
 Hence $3.27 \pm (0.13 \text{ to } 0.14)$
OR (3.13 to 3.14, 3.4(0) to 3.41)
AWFW

A1

6

- (b) Value of 2.90 is **below / outside** CI
ft on (a); OE

B1ft

Suggests **mean height** of adult male
Asian elephants is less **than / different**
from that / mean height
 of adult male **African elephants**
ft on (a); OE

B1ftdep

2

[8]

Q19.

(a) $\bar{v} = \frac{1179}{10} = 117.9$

B1

$$s^2 = \frac{1014.9}{9} = 112.8 \Rightarrow s = 10.6$$

$$\sigma^2 = 101.5 \quad (\sigma = 10.08)$$

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B1

$$t_{0.995} = 3.250$$

B1

99% confidence interval:

$$\text{or use of } \frac{\sqrt{101.5}}{3} = 3.359$$

$$117.9 \pm \frac{10.6}{\sqrt{10}} \times 3.250$$

$$(their \bar{v}) \pm \frac{(their s)}{\sqrt{10}} \times t_9$$

M1

$$= 117.9 \pm 10.9$$

(on $\bar{x}$, s and $t_9 = 3.25$)

A1ft

$$= (106.98, 128.82)$$

$$= (107, 129)$$

AWRT (107, 129)

A1

Assumption:

Speeds form a Normal Distribution

B1

7

- (b) John's claim is unlikely since 130 mph lies outside the confidence interval.

E1

1

[8]

Q20.

$$n = 100 \quad \bar{x} = 1.90 \quad s = 3.32$$

(a) Mean = $60 + \bar{x}$

$$= 61.9$$

CAO

M1

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A1

Standard deviation = 3.32

CAO

B1

3

- (b) 98% $\Rightarrow z = 2.32$ to 2.33
AWFW (2.3263)

($\Rightarrow t = 2.36$ to 2.37)
AWFW (2.364)

B1

CI for μ is $\bar{x} \pm z/t \times \frac{s}{\sqrt{n}}$

Used; must have $\sqrt{n}$ with $n > 1$

M1

Thus $61.9 \pm 2.3263 \times \frac{3.32}{\sqrt{100}}$
 ft on (a) and z/t only

A1ft

Hence $61.9 \pm (0.7 \text{ to } 0.8)$

or $(61.1 \text{ to } 61.2, 62.6 \text{ to } 62.7)$

Accept $1.03 \pm (0.012 \text{ to } 0.013)$

AWFW

Accept $(1.01 \text{ to } 1.02, 1.04 \text{ to } 1.05)$

A1

4

(c) $\bar{s} >> 1 \text{ hour or } 60 \text{ minutes:}$

Not valid as UCL = 1 hour

(Accept Both limits = 1 hour)

Dependent on

UCL = 62.6 to 62.7 or

UCL = 1.04 to 1.05

B1dep

1

EXAM PAPERS PRACTICE [8]

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Q21.

(a) (i)

x:	-5	-3	-1	1	3	5	7	9
F:	4	9	13	27	21	15	7	4

Mean $\left(\frac{\sum x}{n}\right) = 1.9$

CAO (190)

B2

(0.9 to 2.9)

AWFW

(B1)

Standard deviation (s_{n-1} or σ_n) =

(1452)

3.3(0) to 3.32

AWFW (3.31967)

B2

(3(.00) to 3.5(0))

AWFW (3.30303)

(B1)

If no marks scored but $\sum fx$ attempted
and result divided by 100

(M1)

4

(ii) Mean = $60 + \bar{x}$

M1

= 61.9

ft on (a)(i)

A1ft

Standard deviation = 3.3(0) to 3.32

*ft on (a)(i); accept 'same as' only
providing answer in (a)(i)*

B1ft

3

(b) (i) 98% $\Rightarrow z = 2.32$ to 2.33

AWFW (2.3263)

B1

($\Rightarrow t = 2.36$ to 2.37)

AWFW (2.364)

CI for μ is $\bar{x} \pm z/t \times \frac{s_{n-1} \text{ or } \sigma_n}{\sqrt{n} \text{ or } n-1}$

Used; must have $\sqrt{n}$ with $n > 1$

M1

Thus $61.9 \pm 2.3263 \times \frac{3.3 \text{ to } 3.32}{\sqrt{100} \text{ or } 99}$

ft on (a)(ii) and z/t only

A1ft

Hence $61.9 \pm (0.7 \text{ to } 0.8)$

or (61.1 to 61.2, 62.6 to 62.7)

Accept $1.03 \pm (0.012 \text{ to } 0.013)$

AWFW

Accept (1.01 to 1.02, 1.04 to 1.05)

A1

4

- (ii) Mean and SD based upon grouped data
SD (**not mean**) calculated from a sample
CLT used / Times (may) not (be) normal

Actual times/values unknown

Or equivalent

B1

1

- (c) $S > 1$ hour or 60 minutes:

Valid as $74/100$ or 0.74 or $74\% > 50\%$

Must use 74 etc

Or equivalent

B1

$\bar{s} \gg 1$ hour or 60 minutes:

Not valid as $UCL \approx 1$ hour

(Accept Both limits ≈ 1 hour)

Dependent on

$UCL = 62.6 \text{ to } 62.7$ or

$UCL = 1.04 \text{ to } 1.05$

B1dep

2

EXAM PAPERS PRACTICE [14]

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Q22.

(a) (i) $\bar{x} = 3.19$ and $s^2 = \frac{1.849}{9} = 0.2054$

both ($s = 0.453$)

B1

$t_9 = 3.250$

B1

99% confidence interval:

$$3.19 \pm 3.250 \times \frac{\sqrt{0.2054}}{\sqrt{10}}$$

$$3.19 \pm (\text{their } t_9) \times \frac{\sqrt{0.2054}}{\sqrt{10}}$$

M1

$$= 3.19 \pm 0.4658$$

A1ft

$$= (2.72, 3.66)$$

(2.72 to 2.73, 3.65 to 3.66)

A1

5

- (ii) Reasonable claim, with 3.5 within the
99% confidence interval
dep on correct CI in (a)(i)

B1E1

2

(b) $0.01 \times 200 = 2$

B1

1

[8]

Q23.

(a) $90\% \Rightarrow z = 1.64 \text{ to } 1.65$
AWFW (1.6449)

B1

or
 $90\% \Rightarrow t = 1.66 \text{ to } 1.67$
AWFW (1.6649)

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(B1)

(Knowledge of the t -distribution is **not**
required in this unit)

CI for μ is
$$\bar{x} \pm (z \text{ or } t) \times \frac{(s_{x-1} \text{ or } s_x)}{\sqrt{n}}$$

Used; must have $\sqrt{n}$ with $n > 1$

M1

Thus

$$184 \pm (1.6449 \text{ or } 1.6649) \times \frac{(32 \text{ or } 32.2)}{(\sqrt{78} \text{ or } \sqrt{77})}$$

ft on z or t only

A1ft

Hence $184 \pm (5.94 \text{ to } 6.13)$

or $\pounds 184 \pm \pounds 6$

or ($\pounds 178, \pounds 190$)

AWRT; ignore units

A1

4

- (b) (i) **Likely to be valid**
Accept 'valid' or equivalent

B1

- (ii) Different plays have different:
programme prices, sales, marketing, etc
theatre or audience sizes, etc
popularity, artists, etc

B1

so

↑Dep↑

Unlikely to be valid

Accept 'not valid' or equivalent

B1

3

[7]

Q24.

- (a) (i) $P(X < 40) = P\left(Z < \frac{40 - 38}{5}\right) =$
Standardising (39.5, 40 or 40.5) with 38
and ($\sqrt{5}$, 5 or 5^2) and/or $(38 - x)$

M1

$$P(Z < 0.4) = 0.655 \text{ to } 0.66$$

AWFW (0.65542)

A1

2

- (ii) $P(30 < X < 40) =$
 $P(X < 40) - P(X < 30) =$
(i) $- P(X < 30) =$
(i) $- P(Z < -1.6) =$
Difference or equivalent
Standardising other than 40 and 30 $\Rightarrow$

M1

$$\begin{aligned}
 & (i) - \{1 - P(Z < +1.6)\} = \\
 & 0.65542 - \{1 - 0.94520\} = \\
 & \text{Area change}
 \end{aligned}$$

m1

$$\begin{aligned}
 & 0.6 \text{ to } 0.601 \\
 & \text{AWFW (0.60062)}
 \end{aligned}$$

A1

3

$$\begin{aligned}
 (iii) \quad & 75\% (0.75) \Rightarrow z = \pm 0.674 \text{ to } \pm 0.675 \\
 & \text{AWFW (0.6745)}
 \end{aligned}$$

B1

$$z = \frac{x - 38}{5}$$

Standardising x with 38 and 5

M1

$$= -0.6745$$

Equating z -term to z -value but not using
 $0.75, 0.25, |1 - z|$ or $\Phi(0.75) = 0.77337$

m1

$$x = 34.6 \text{ to } 35$$

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A1

4

$$\begin{aligned}
 (b) \quad (i) \quad & 98\% \Rightarrow z = 2.32 \text{ to } 2.33 \\
 & \text{AWFW (2.3263)}
 \end{aligned}$$

B1

or

$$\begin{aligned}
 & 98\% \Rightarrow t = 2.42 \text{ to } 2.43 \\
 & \text{AWFW (2.426)}
 \end{aligned}$$

(B1)

(Knowledge of the t -distribution is not required in this unit)

CI for μ is $\bar{y} \pm (z \text{ or } t) \times \frac{(s_{n-1} \text{ or } s_n)}{\sqrt{n}}$
 Used; must have $\sqrt{n}$ with $n > 1$

M1

Note that $19.1 \times \sqrt{\frac{40}{39}} = 19.34332$

$19.1 \times \sqrt{\frac{40}{39}} = 19.58974$
 Max of B1 M1 A0ft A1

Thus

$107 \pm (2.32 \text{ to } 2.43) \times \frac{(19.1 \text{ or } 19.3 \text{ to } 19.4)}{(\sqrt{40} \text{ or } \sqrt{39})}$

ft on z or t only

A1ft

Hence $107 \pm (7.00 \text{ to } 7.55)$

ie $107 \pm (7 \text{ to } 8)$
 or $(99 \text{ to } 100, 114 \text{ to } 115)$

AWFW

A1

4

(ii) $2\frac{1}{2} \times (\text{mean for adult males eels})$
 $= 2.5 \times 38 = 95$
 CAO

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B1
 ↑Dep↑

Since $95 < \text{LCL}$
 OE; ft on CI

B1ft
 ↑Dep↑

Claim appears valid
 OE; ft on CI

B1ft

3

[16]

Q25.

(a) $95\% \Rightarrow z = 1.96$

CAO

B1

or

$$95\% \Rightarrow t = 2.0 \text{ to } 2.01$$

AWFW (2.009)

(B1)

(Knowledge of the t -distribution is not required in this unit)

CI for μ is $\bar{x} \pm (z \text{ or } t) \times \frac{(s_{n-1} \text{ or } s_n)}{\sqrt{n}}$

Used; must have $\sqrt{n}$ with $n > 1$

M1

Note that

$$25.1 \times \sqrt{\frac{50}{49}} = 25.35483$$

$$25.1 \times \frac{50}{49} = 25.61224$$

Max of B1 M1 A0ft A1

Thus

$$234 \pm (1.96 \text{ or } 2.009) \times \frac{(25.1 \text{ or } 25.3 \text{ to } 25.4)}{(\sqrt{50} \text{ or } \sqrt{49})}$$

ft on z or t only

A1ft

Hence $234 \pm (6.95 \text{ to } 7.30)$

ie 234 ± 7

or $(227, 241)$

AWRT

A1

4

- (b) Customers are likely to choose large / similar sized potatoes

OE; accept any sensible alternative

B1

1

[5]

Q26.

- (a) Assumption that the speeds of the cars passing through the village are normally distributed

B1

$$\bar{x} = 35.6$$

B1

$$s^2 = 38.27 \quad (s = 6.186)$$

$$(\sigma^2 = 34.44 \quad (\sigma = 5.869))$$

B1

99% Confidence Interval for μ

$$= 35.6 \pm 3.250 \times \frac{6.186}{\sqrt{10}}$$

or use of $\frac{\sqrt{34.44}}{3}$

B1

$$= 35.6 \pm 6.36$$

on their mean and standard deviation

M1A1ft

$$= (29.2, 42.0)$$

CAO (29.24, 41.96)

A1

7

(b) Confidence interval includes 30 mph

B1ft

80% of sample exceed 30 mph limit

B1

Speed limit **not** adhered to

dependent on previous B1

B1

3

[10]

Q27.

(a) Mean = $\frac{286.5}{50} = 5.73$

CAO

B1



$$\text{Standard deviation} = \sqrt{\frac{45.16}{49 \text{ or } 50}} = 0.95 \text{ to } 0.961$$

AWFW

B1

2

(b) $99\% \Rightarrow z = 2.57 \text{ to } 2.58$

AWFW

2.5758

B1

CI for μ is $\bar{x} \pm z \times \frac{(\sigma \text{ or } s)}{\sqrt{n}}$

Use of

Must have $\div (\sqrt{n})$ with $n > 1$

M1

Thus $5.73 \pm 2.5758 \times \frac{(0.95 \text{ to } 0.961)}{\sqrt{50}}$

ft on z and $S^2/\bar{x} > 0$ but not on x Accept only 50 or 49 for n

A1ft

$$5.73 \pm (0.34 \text{ to } 0.36)$$

↑

$$(5.37 \text{ to } 5.39, 6.07 \text{ to } 6.09)$$

AWFW

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A1

4

(c) CI excludes both values of 5 and $6\frac{1}{2}$

ft on(b); OE

B1ft

so

↑

Neither claim appears valid

*Dependent**ft on(b); OE*

B1ft

or

CI excludes 5 so claim not valid

ft on(b); OE

(B1ft)

and

CI excludes $6\frac{1}{2}$ so claim not valid

ft on(b); OE

(B1ft)

2

[8]

Q28.

(a) $\bar{x} = 301.0$

AWFW 24.4 to 24.5

B1

(b) $S^2 = (4.945)^2 = 24.5$

B1

2

$v = 9$

B1B1

$t = 3.250$

99% confidence interval for μ

$$= 301.0 \pm 3.250 \times \left(\frac{4.945}{\sqrt{10}} \right)$$

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$$(\text{their } \bar{x}) \pm t \times \left(\frac{\text{their } \sigma}{\sqrt{10}} \right)$$

M1

$= 301.0 \pm 5.08$

A1

$= (295.9, 306.1)$

A1

5

- (c) Since 290 is less than the lower limit of the confidence interval, it is probably safe to assume that the mean weight of packets of rice exceeds 290 grams

E2

2

[9]

Q29.

(a) $\bar{x} = 8.0$

B1

$S = 2.121$

B1

$V = 8$

B1

$t = 1.860$

(on their v)

B1ft

90% confidence interval for μ

$= 8 \pm 1.860 \left(\frac{2.121}{3} \right)$

M1

$= 8 \pm 1.315$

A1ft

$= (6.68, 9.32)$

(6.68 to 6.69, 9.31 to 9.32)

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A1

7

(b) The Headteacher's claim seems to be slightly optimistic

*Headteacher's claim isn't supported by the evidence **and***

E1ft

because value of 5 outside the confidence interval

[It appears that the mean time to see a mathematics teacher is greater than 5 minutes]

E1ft

2

[9]

Q30.

(a) Weight, $X \sim N(1012, 5^2)$



$$(i) \quad P(X < 1015) = P\left(Z < \frac{1015 - 1012}{5}\right)$$

standardising (1014.5, 1015 or 1015.5) with 1012 and
($\sqrt{5}$, 5 or 5^2) and/or (1012 – x)

M1

$$= P(Z < 0.6)$$

CAO; ignore sign

A1

$$= \Phi(0.6) = 0.725 \text{ to } 0.726$$

AWFW (0.72575)

A1

3

$$(ii) \quad P(X > 1005) = P\left(Z > \frac{1005 - 1012}{5}\right)$$

standardising (1004.5, 1005 or 1005.5)

with 1012 and ($\sqrt{5}$, 5 or 5^2) and/or (1012 – x)

M1

$$= P(Z > -1.4) = P(Z < 1.4)$$

area change

m1

$$= 0.919 \text{ to } 0.920$$

AWRT (0.91924)

A1

3

$$(iii) \quad P(1005 < X < 1015) = (i) - [1 - (ii)]$$

or equivalent

M1

$$= 0.72575 - [1 - 0.91924]$$

$$= 0.644 \text{ to } 0.646$$

AWFW (0.64499)

ft on (i) and (ii) providing > 0

A1ft

- (b) (i) Mean, $\bar{y} = 502.85$
CAO; stated or implied

B1

Variance, $\sigma^2 / s^2 = 50.3$ to 51.5
or
Standard deviation, $\sigma / s = 7.09$ to 7.17
AWFW; either stated or implied

B1

98% $\Rightarrow z = 2.32$ to 2.33
AWFW (2.3263)

B1

or
98% $\Rightarrow t = 2.40$

B1

(Knowledge of the t -distribution is not required in this unit)
AWRT (2.403)

B1

CI for μ is $\bar{y} \pm (z \text{ or } t) \times \frac{(\sigma \text{ or } s)}{\sqrt{n}}$
use of; must have $(\pm \sqrt{n})$ with $n > 1$

M1

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$$502.85 \pm (2.33 \text{ or } 2.40) \times \left(\frac{7.09 \text{ to } 7.17}{\sqrt{49 \text{ or } 50}} \right)$$

ft on $\bar{x}$, z/t and $\sigma/s > 0$;
 $n = 49$ or 50 only

A1ft

or
 $502.85 \pm (2.32 \text{ to } 2.46)$
or
(500.3 to 500.6, 505.1 to 505.4)
AWFW; dependent on A1

A1 dep

- (ii) **Given sample:**
6 in 50/some of packets have weights below 500 grams
or equivalent

B1

Confidence interval:

CI > 500

ft on CI in (b)(i)

B1ft

Conclusion:

Statement does not appear justified

or equivalent.

dependent on both B1 and B1ft

B1 dep

3

[17]

Q31.

- (a) (i) Mean, $\bar{x} = 505.2$
CAO; stated or implied

B1

99% $\Rightarrow z = 2.57$ to 2.58
AWFW (2.5758)

B1

Or
99% $\Rightarrow t = 3.25$
AWRT (3.250)

B1

(Knowledge of the t -distribution is not required in this unit)

CI for μ is $\bar{x} \pm (z \text{ or } t) \times \frac{(\sigma \text{ or } s)}{\sqrt{n}}$
use of; must have $(\pm \sqrt{n})$ with $n > 1$

M1

Thus $505.2 \pm 2.5758 \times \frac{6}{\sqrt{10}}$
ft on $\bar{x}$ and z only

A1ft

or

$$505.2 \pm 3.25 \times \left(\frac{5.96}{\sqrt{10}} \text{ or } \frac{5.65}{\sqrt{9}} \right)$$

ft on $\bar{x}$ only

A1ft

Hence

$$505.2 \pm 4.9$$

use of $t \Rightarrow 505.2 \pm 6.1$

or
(500.3, 510.1)
AWRT

A1

5

- (ii) Given sample:
3 in 10/ some of packets have weights below 500 grams
or equivalent

B1

Confidence interval:
CI > 500
ft on CI in (a)(i)

B1ft

Conclusion:
Statement does not appear justified
or equivalent
dependent on both B1 and B1ft

B1 dep

3

- (b) 0.01 or 1%
CAO; or equivalent

B1

1

[9]

Q32.

- (a) Volume $\sim N(\mu, 3.5^2)$

Mean,

$$\bar{x} = \frac{1830}{12} = 152.5$$

CAO ($s_{n-1} = 3.778, s_n = 3.617$)

B1

98% $\Rightarrow z = 2.3263$

AWFW 2.32 to 2.33

B1

CI for μ is $\bar{x} \pm z \times \frac{(\sigma \text{ or } s)}{\sqrt{n}}$

Use of

Must have $(\div \sqrt{n})$ with $n > 1$

M1

Thus $152.5 \pm 2.3263 \times \frac{3.5}{\sqrt{12}}$

ft on $\bar{x}$ and z only

A1ft

(150.1 to 150.2, 154.8 to 154.9)

AWFW

A1

5

- (b) Evidence, from CI, that mean volume is (above) 150 ml

ft on CI in part (a); must be clear comparison of mean of 150 with CI

B1ft

In sample, some cans have volumes less than 150 ml

Or reference to range of can volumes in sample

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Thus claim of 150 ml is not justified

Dependent upon making some comment about mean volume and some comment about individual can volume or range of can volumes

B1dep

3

[8]

Q33.

- (a) (i) Time, $X \sim N(12, 2.5^2)$

$P(X < 15) = P\left(Z < \frac{15 - 12}{2.5}\right)$

standardising (14.5, 15 or 15.5) with

$(\sqrt{2.5}, 2.5 \text{ or } 2.5^2) \text{ and/or } (12 - x)$

M1

$$P(Z < 1.2) = 0.885$$

AWRT (0.88493)

A1

2

(ii) $P(10 < X < 15) = (i) - P(X < 10)$

OE

M1

$$= 0.88493 - P(Z < 0.8)$$

$$= 0.88493 - (1 - \Phi(0.8))$$

area change

$$= 0.88493 - (1 - 0.78814) = 0.673$$

AWRT (0.67307)

A1

3

(b) (i) $\bar{y} = \frac{835.0}{50} = 16.7$

CAO

B1

$$s^2 = \frac{533.61}{49} = 10.89 \quad \text{or} \quad s = 3.3$$

CAO; either

$$v = \frac{533.61}{50} = 10.6722 \quad \text{or} \quad \sqrt{v} = 3.2668$$

AWRT 10.67 or AWRT 3.27

B1

$$99\% \Rightarrow z = 2.5758$$

AWFW 2.57 to 2.58

B1

CI for μ is

$$\bar{y} \pm z \times \frac{(s \text{ or } \sqrt{v})}{\sqrt{n}}$$

use of; must have ($\div \sqrt{n}$) with $n > 1$

M1

Thus: $16.7 \pm 2.5758 \times \frac{\sqrt{50}}{\sqrt{2}}$
ft on $\bar{y}, z, (s \text{ or } \sqrt{v})$; not on n

A1ft

Thus: (15.5, 17.9)
AWRT; dependent on $\div 49$ for variance unless subsequently corrected

A1

6

- (ii) Adding 25% to 12 gives 15
CAO; seen somewhere

B1

Since 15 is outside/below CI
ft on (b)(i); must use 15

E1ft

Mustafa's suspicion is supported
ft on (b)(i); must use 15

B1ft

3

[14]

Q34.

- (a) (i) $\bar{x} = 179.6$
CAO

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B3

3

$s_x^2 = 133.16$
133 to 133.2
(B1 M1 A1 if by formula)

- (ii) $v = 10 - 1 = 9$

B1

Critical value of $t = 1.833$

B1

90% confidence limits for μ_x are

$$179.6 \pm 1.833 \sqrt{\frac{133.16}{10}}$$

(M1 only if z-value used)

M1

giving (172.9, 186.3)
ft on answers to (i) and t value

A1ft

AWFW (172.9 to 173, 186 to 186.3)

A1

5

- (b) (i) Confidence limit for price in £ is

$$\left(1.20 \times \frac{173}{1000}, 1.20 \times \frac{186}{1000} \right)$$

M1

$$= (0.21, 0.22)$$

AWRT; or (21p, 22p); ft on CI in (a)(ii)

A1ft

2

- (ii) 20 pence is below confidence interval
so oranges are cheaper/ less profit per orange

E1ft

comparing 20p with CI; ft on CI
sensible deduction
(both marks if lower end of CI implied)

E1

2

[12]

Q35.

(a) $\text{Var}(T) = s^2 = \frac{279.8929}{49} = 5.71$
awrt (5.7121)

B1

$$\text{SE}(\bar{T}) = \sqrt{\frac{\text{Var}(T)}{50}}$$

use of

M1

= 0.338

Awrt [cannot be scored in part (b)(i)]

A1

3

(b) (i) $\bar{x} = \frac{143.5}{50} = 2.87$

cao; can be scored in part (a)

B1

99% implies $z = 2.5758$

awfw 2.57 to 2.58

B1

CI for μ is: $\bar{x} \pm z \times \frac{(s \text{ or } \sigma)}{\sqrt{n}}$
use of; must have $\sqrt{n}$ with $n > 1$ or equivalent

M1

or $\bar{x} \pm z \times SE(\bar{x})$
or $\sqrt{n}$ in $SE(\bar{x})$

Thus: $2.87 \pm (2.5758 \times 0.338)$

f.t. on $\bar{x}$, z and $SE(\bar{x}) > 0$; accept

$\bar{x} = 143.5$ only if clearly stated

A1f.t.

Thus: (2.00, 3.74)
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awrt; accept 2

dependent on $\div$ by 49 in part (a) unless subsequently corrected

A1

5

(ii) Evidence to suggest that $\mu = 3.5$

f.t. on part (b)(i)

B1f.t.

as 3.5 inside CI

clearly stated; f.t. on part (b)(i)

B1f.t.

2

(c) Now evidence (to suggest that μ has changed/increased from 3.5 (as 3.5 outside/below CI)

reason not required

B1

Also evidence (to suggest μ has increased during three months)
as CIs do not overlap

reason not required

B1

2

[12]

Q36.

- (a) Mean, $\bar{x} = 413$
CAO

B1

99% $\Rightarrow z = 2.5758$
AWFW 2.57–2.58

B1

CI for μ is $\bar{x} \pm z \times \frac{\sigma}{\sqrt{n}}$
use of, must have $\sqrt{n}, n \neq 1$
use of unbiased estimate M1A0A0

M1

$\therefore 413 \pm 2.5758 \times \frac{3}{\sqrt{12}}$
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f.t. on $\bar{x}$ and z, allow 3^2 or $\sqrt{3}$ but must be $\sqrt{12}$

A1f.t.

$\therefore 413 \pm 2.2$
 $\therefore (410.8, 415.2)$
AWRT Incorrect z $\Rightarrow$ B1BO M1A1f.t.
A0 max

A1

5

- (b) As 99% CI excludes 410
f.t.on (a)

B1f.t.

Unlike to be correct
f.t.on (a), dependent on B1

Q37.

(a) $A \sim N(10.3, 0.16^2)$

(i) $P(A < 10.5) = P\left(Z < \frac{10.5 - 10.3}{0.16}\right)$
*standardising 10.45, 10.5, 10.55 with σ
 σ^2 , $\sqrt{\sigma}$ and/or $(\mu - x)$*

M1

$= P(Z < 1.25) =$
CAO ± 1.25

A1

0.89 (435)
AWRT

A1

3

(ii) $P(A > a) = 0.75$
 $\Rightarrow z = -0.6745$
*AWFW ($\pm$)0.674 to ± 0.675
(not 0.67 or 0.68)*

B1

$\frac{a - 10.3}{0.16} = z$
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standardising value with σ , σ^2 , $\sqrt{\sigma}$ and/or $(\mu - x)$

M1

$\Rightarrow \frac{a - 10.3}{0.16} = -0.6745$

equating two z-values (m0 for using 1-z-value)

m1

$a = 10.2$
AWRT

A1

4

(b) (i) $\hat{\mu} = \bar{x} = 24.9$

CAO
(B0 for bias correction)

B1

1

- (ii) $SE(\hat{\mu}) = \frac{\sigma}{\sqrt{n}} =$
 use of $(\sigma, \sigma^2, \sqrt{\sigma}) \div (n, \sqrt{n})$ with $n > 1$
 (accept with $0.16, 0.16^2, \sqrt{0.16}$)
 (M0 for bias correction)

M1

0.02
 Cannot gain (i) and (ii) from (iii)
 CAO

A1

2

- (iii) 95% $\Rightarrow z = 1.96$
 CAO

B1

CI for μ is: $\hat{\mu} \pm z \times SE(\hat{\mu})$
 use of; must have used $n > 1$ in (ii)

M1

$$\therefore 24.9 \pm 1.96 \times 0.02$$

$$\therefore 24.9 \pm 0.04$$

f.t. on (i) and (ii)

A1f.t.

24.86, 24.94

AWRT

A1

4

- (iv) As 95% CI excludes 25
 f.t. on (iii)

B1f.t.

Disagree with claim
 f.t. on (iii); dep on B1f.t.

E1f.t.

2

Only mark if CI of the form (a,b) with a and b determined

[16]

Q38.

(a) (i) $\hat{\mu} = \bar{x} = \frac{\sum x}{25} = \frac{88.2}{25}$
use of

M1

$= 3.528$
CAO; accept 3.53

A1

$\sigma^2 = s^2 = \frac{1}{24} \left(\sum x^2 - \frac{(\sum x)^2}{25} \right) =$
use of
[M0 for $\div 25$]

M1

$\frac{1}{24} \left(326.2 - \frac{88.2^2}{25} \right) =$
0.625 to 0.627
AWFW

A1

4

(ii) CI for μ is:

$\bar{x} \pm \left(\frac{t}{z} \right) \frac{s}{\sqrt{n}}$

use of

M1

$t_{24}(0.995) = 2.797$
CAO; accept 2.80
($z = 2.576$)

B1

$\therefore 3.528 \pm 2.797 \sqrt{\frac{0.626}{25}}$
f.t. on (i), (ii) & $\frac{t}{z}$

A1f.t.

$\therefore 3.53 \pm 0.44$

∴ (3.09, 3.97)

AWRT

A1

4

- (b) As 99% CI **does not include** 4.1

f.t. on CI

B1f.t.

Mean weight of babies born in hospital is
different from/less than that in region

f.t.on CI; dep on B1f.t.

E1f.t.

2

[10]

Q39.

- (a) $\bar{x} = 4256/400 = 10.64$

10.64 allow 10.6

B1

95% confidence interval for mean

1.96

B1

$$10.64 \pm 1.96 \times \frac{3.68}{\sqrt{400}}$$

Use of $\frac{3.68}{\sqrt{400}}$, allow use of $3.68 \sqrt{\frac{400}{399}}$

M1

$$10.64 \pm 0.361$$

*Correct method for interval, their mean-
allow incorrect z-value*

m1

(10.28 , 11.00)

*10.28 (10.275 ~ 10.3) and
11.00 (10.995 ~ 11.005)
or 10.64 ± 0.361 (0.36 ~ 0.361)*

A1

5

- (b) (i) $E(X) = 5 \times 0.15 + 10 \times 0.63 + 15 \times 0.15 + 20 \times 0.07$

Method for E(X)

M1

$$= £10.7$$

10.7 cao – ignore units

A1

(ii) $E(X^2) = 25 \times 0.15 \times 100 \times 0.63 + 225 \times 0.15 + 400 \times 0.07 = 128.5$

Method for $E(X^2)$ may be implied

M1

Method for s.d., their answers to (i) and (ii) – allow variance if called variance

M1

(iii) s.d. of $X = \sqrt{128.5 - 10.7^2} = £3.74$

*Completely correct method for s.d. –
Allow variance if called variance*

m1

3.74 (3.74 ~ 3.745)

A1

6

- (c) mean within confidence interval
calculated in (a), standard deviation close
to observed standard deviation. Model
appears plausible.

ft Mean within confidence interval –

E1ft

allow similar

ft s.d. similar to observed

Correct conclusion based on correct calculations

E1ft

3

- (d) (i) $\bar{x} = 2342/200 = 11.71$
11.71 or 11.7

B1

$$H_0 \mu = 11.00$$

One correct hypothesis – generous

B1

$$H_1 \mu > 11.00$$

Both correct – ungenerous

B1

$$z = \frac{11.71 - 11.00}{\frac{3.42}{\sqrt{200}}} = 2.94$$

M1

Use of $\frac{3.42}{\sqrt{200}}$, allow use of $3.42 \sqrt{\frac{200}{199}}$

M1

Correct method for z, ignore sign

m1

2.94 (2.93 ~ 2.94)

A1

critical value 1.6449

1.6449 or 1.645 or 1.64 or 1.65
allow t = 1.652 or 1.653

B1

Reject H_0 significant evidence mean exceeds

A1ft

8

Correct conclusion, their figures, must be compared with appropriate tail of z.
needs previous

M1

- (ii) Since £11 was upper limit of confidence interval for mean, there is strong evidence that the mean has increased

Evidence sales have increased

E1

Since £11 upper limit of confidence interval

E1

2

- (iii) Have total sales of petrol increased? How much does the scheme cost? Have other sales increased? Etc

Any sensible point

E1

A second sensible point

E1

2

[26]

Q40.

(a) $x = \frac{4256}{400} = 10.64$
10.64 allow 10

B1

95% confidence interval for mean

$$10.64 \pm 1.96 \times \frac{3.68}{\sqrt{400}}$$

1.96

B1

use of 3.68/ $\sqrt{400}$

M1

$$10.64 \pm 0.361$$

*correct method for interval – their mean -
allow incorrect z-value*

m1

$$(10.28, 11.00)$$

*10.28 (10.275 ~ 10.3) and
11.00 (10.995 ~ 11.005)
or 10.64 $\pm$ 0.361 (0.36 ~ 0.361)*

A1

5

(b) (i) $x = \frac{2342}{200} = 11.71$
95% confidence interval for mean

$$11.71 \pm 1.96 \times \frac{3.42}{\sqrt{200}}$$

completely correct method

M1

$$11.71 \pm 0.474$$

$$(11.24, 12.18)$$

*11.24 (11.2 ~ 11.3) and 12.18 (12.15 ~ 12.2)
or 11.71 (11.7 ~ 11.71) $\pm$ 0.474 (0.473 ~ 0.475)*

A1

2

- (ii) Since confidence intervals for mean
before and after the offer do not overlap

there is strong evidence that the mean has increased

confidence intervals do not overlap

E1ft

correct conclusion based on correct calculation and reason

E1

2

- (iii) Have **total** sales of petrol increased?
How much does the scheme cost? Have other sales increased? Etc

Any sensible point

E1

A second sensible point

E1

2

[11]

Q41.

(a) $\bar{x} = 135.6$

135.6 cao

B1

$s = 0.45552$

0.45552 (0.455 to 0.456) may be implied by correct interval

B1

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95% confidence interval for mean is $135.6 \pm 2.306 \times \frac{s}{\sqrt{9}}$

8df

B1

2.306

B1f.t.

Use of their $\frac{s}{\sqrt{9}}$

i.e. 135.6 ± 0.350

135.25 to 135.95

Correct method for interval

m1

Answer given to 4,5 or 6 sf, must be an interval

B1

135.25 (135.2 to 135.3) and
 135.95 (135.9 to 136)
Or 135.6 **cao** ± 0.35 (0.349 to 0.351)

A1

8

(b) $135.8 \pm 1.96 \times \frac{3.9}{\sqrt{60}}$
1.96 cao

B1

Use of $\frac{3.9}{\sqrt{60}}$

B1

135.8 ± 0.987

Correct method for interval

m1

134.81 to 136.79

134.81 (134.8 to 134.820) and 136.79 (136.78 to 136.8)
Or 135.8 **cao** ± 0.987 (0.98 to 1)

A1

4

(c) (i) Decrease

Decrease

B1

1

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(ii) Increase - narrower interval $\rightarrow$ more likely to identify need for overhaul if $\mu \neq 135.0$

Increase

E1

Reason

E1

2

(d) Large variability as in (b) will lead to unsatisfactory production even if mean on target. Might prefer small variability as in (a) even if mean slightly off target.

Large variability unsatisfactory/makes deviation from 135 difficult to detect.

E1

Small variability with mean slightly off target may be preferable

E1

2

[17]

Q42.

(a) $\bar{x} = 135.6$

135.6 cao

B1

95% confidence interval for mean is

$$135.6 \pm 1.96 \times \frac{0.42}{\sqrt{9}}$$

1.96

B1

Use of $\frac{0.42}{\sqrt{9}}$

M1

i.e. 135.6 ± 0.274
135.33 to 135.87

Correct method for interval

m1

Answer given to 4, 5 or 6 sf

A1

*135.33 (135.33 to 135.4) and
135.87 (135.8 to 135.9)*

Or 135.6 cao ± 0.274 (0.274 to 0.275)

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A1

6

(b) $135.8 \pm 1.96 \times \frac{3.9}{\sqrt{60}}$

Correct method - their z

M1

135.8 ± 0.987
134.81 to 136.79

*134.81 (134.8 to 134.820) and
136.79 (136.78 to 136.8)*

Or 135.8 cao ± 0.987 (0.98 to 1)

A1

2

(c) (i) Decrease

B1

1

- (ii) Increase - narrower interval → more

E1

likely to identify need for overhaul if $\mu \neq 135.0$.

E1

2

- (d) Large variability as in (b) will lead to unsatisfactory production even if mean on target. Might prefer small variability as in (a) even if mean slightly off target.

Large variability unsatisfactory/makes deviation from 135 difficult to detect.

E1

Small variability with mean slightly off target may be preferable

E1

2

[13]

Q43.

- (a) $\bar{x} = 48.4$ $s = 18.094$
48.4 cao
18.094 (18.09 – 18.1)

99% confidence interval

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B1 B1

M1

$$48.4 \pm 3.250 \times \frac{18.094}{\sqrt{10}}$$

9 df
3.250 – their df

B1 B1ft

ie 48.4 ± 18.6

29.8 – 67.0

*Use of their $\frac{s.d.}{\sqrt{10}}$
Completely correct method – their t
29.8(29.75 – 29.85)*

and 67.0 (66.95 – 67.05)
 or 48.4 and 18.6 (18.55 – 18.65)
 sc B1 B1 M1 M1 available if z used

M1 m1 A1

8

- (b) (i) 95% confidence interval
 1.96 – allow t 1.979 – 1.984

B1

$$49.5 \pm 1.96 \times \frac{16.5}{\sqrt{110}}$$

Use of $\frac{16.5}{\sqrt{110}}$
 Their z

$$49.5 \pm 3.08$$

M1

A1

4

$$46.42 - 52.58$$

$49.5 \pm 3.08(3.08 - 3.09)$
 or 46.42 (46.1 – 46.42)
 and 52.58(52.58 – 52.59)
 Allow 46.4 and 52.6

- (ii) $49.5 \pm 1.96 \times 16.5$
 Their z

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M1

$$17.2 - 81.8$$

$49.5 \pm 32.34(32.3 - 32.4)$
 or 17.2 (17.1 – 17.2) and
 81.8 (81.8 – 81.9)

A1

2

- (c) Mean greater than 25 but many
 Mean $\neq$ 25

E1

individuals will be less than 25
 Mean > 25
 Some individuals < 25

E1 E1

3

- (d) Interval narrower
Interval narrower

E1

No need for normal assumption

Normal not needed

Or better estimate of σ , more representative, more reliable **not** more accurate

E1

2

- (e) Both samples random so valid suggestion
valid

E1

Would be better but not much

$$\frac{18.5}{\sqrt{110}} \text{ very similar to } \frac{18.5}{\sqrt{120}}$$

Better

E1

Not much - any three points

E1

3

EXAM PAPERS PRACTICE [22]

Q44. ©2025 Exam Papers Practice. All rights reserved.

- (a) $\bar{x} = 48.4$

48.4 cao

B1

99% confidence interval

$$48.4 \pm 2.5758 \times \frac{14.8}{\sqrt{10}}$$

$$2.5758(2.57 - 2.58)$$

$$\text{Use of } \frac{14.8}{\sqrt{10}} \text{ or } \frac{5}{\sqrt{10}}$$

B1 M1

ie 48.4 ± 12.1

m1

36.3 – 60.5

*Completely correct method – candidate's
36.3 (36.3 – 36.4) and 60.5 (60.4 – 60.5)
or 48.4 and 12.1 (12 – 12.1)*

A1

5

(b) (i) 95% confidence interval

$$49.5 \pm 1.96 \times \frac{16.5}{\sqrt{110}}$$

1.96
Method – their z

B1 M1

$$49.5 \pm 3.08$$

*49.5 ± 3.08 (3.08 – 3.09)
or 46.42 (46.41 – 46.42)
and 52.58 (52.58 – 52.59)
Allow 46.4 and 52.6*

A1

3

(ii) $49.5 \pm 1.96 \times 16.5$
Method – their z

M1

$$17.2 - 81.8$$

*49.5 ± 32.34 (32.3 – 32.4)
or ± 17.2 (17.1 – 17.2)
and 81.8 (81.8 – 81.9)*

A1

2

(c) Mean greater than 25 but many
Mean ≠ 25

E1

individuals will be less than 25
Mean > 25

E1

Some individuals < 25

E1

3

(d) Interval narrower

Interval narrower

E1

No need for normal assumption

Normal not needed

E1

2

(e) Both samples random so valid suggestion.

Valid

E1

Would be better but not much ($\frac{49.5}{\sqrt{110}}$)

Better

E1

very similar to ($\frac{49.5}{\sqrt{120}}$)

Not much

Any three points

E1

3

[18]

Q45.

(a) $\bar{x} = 627$

cao

B1

$s = 640.38$

640 ~ 641

B1

95% confidence interval for mean is

9 df

B1

$627 \pm 2.262 \times 640.38 / \sqrt{10}$

for 2.262 or 2.26

attempt to use $s\sqrt{10}$

use of t-distribution

completely correct method—

allow incorrect t value

B1 M1 M1 M1

627 ± 458

$169 \sim 1085$

$457 \sim 459$

or 168 ~ 170 and 1080 ~ 1090

A1

8

- (b) (i) 3 people drank no water

B1

suggesting a mode of zero which is a long way from the mean / not symmetrical

B1

- (ii) amount cannot be negative

B1

3

- (c) (i) $960 \pm 1.6449 \times 240 / \sqrt{135}$

for 1.6449 or 1.645 or 1.64 or 1.65 or 1.655 ~ 1.657

for use of $240 / \sqrt{135}$

for completely correct method

(allow use of t-distribution)

B1 M1 M1

960 ± 34.0

$33.5 \sim 34.5$

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A1

4

$926 \sim 994$

$925.5 \sim 926.5$ and $993.5 \sim 994.5$

- (ii) first day/night shift unlikely to be typical of all employees

B1

1

[16]

Examiner reports

Q1.

In part (a), the majority of students scored at least three marks. The most common error was to incorrectly calculate the standard error by using 10 as the standard deviation rather than the variance. Some students obtained a probability for the z value or found the z value for a different confidence interval. A small minority of students transposed the digits for the sample mean, obtaining 34.5 instead of 35.4. Some students lost the final mark for not giving their confidence interval to four significant figures.

Most students did not score the mark for part (b). Many understood that Dante would reject the null hypothesis but many did not give a reason or were unclear in their explanation. It was common for students to state that 'it' was not in the confidence interval without explaining what they meant by 'it'. A minority of students wrote that Dante would accept the null hypothesis as the sample mean was in the confidence interval. Some students attempted to calculate the probability for the hypothesis test but none were successful.

Q3.

The calculation part was handled well, apart from by those students who used a z -value, and others who ignored the instruction to give the answer to two decimal places. A fair proportion of students recognised the uncertainty introduced because the 2011 mean lay within the 2012 confidence interval, and correctly deduced that the 2012 mean might lie below or above the previous year's mean. Others were far too definite in their conclusions. Strangely, a sizeable proportion of students equated a higher time with a better performance. Very few observed that success in races is not directly linked to mean time and depends also on other athletes' performances.

Q4.

This question usually got students off to a good start with most finding the confidence interval correctly in part (a). Occasional errors were seen with the use of an incorrect z -value or the omission of $\sqrt{25}$. In part (b), despite similar requests appearing on many previous papers, students often did not make an explicit comparison of the stated 20 kg claim with their confidence interval. The use of "it" or "mean" was not accepted nor, of course, was a comparison of 20 kg with 19.9 kg.

Q5.

Most students found the confidence interval correctly. Occasional errors, in addition to simple numerical ones, were the use of $\sum x$ rather than $\bar{x}$, an incorrect z -value or the omission of $\sqrt{25}$. Despite similar requests appearing on many previous papers, students often did not make an explicit comparison of the stated 20 kg claim with their confidence interval. The use of "it" or "mean" was not accepted nor, of course, was a comparison of 20 kg with $\bar{x}$. Again, despite similar questions appearing on previous papers, many students remained unable to distinguish between the two structures in part (b). Thus

0.35

standardising using $\frac{0.35}{\sqrt{10}}$ in both parts was common as was the failure to raise a probability to the power 10 in part (b)(ii). Nevertheless, a significant proportion of students, not all of whom were necessarily high achievers, scored full marks in part (b).

Q6.

The calculation part was well handled, apart from those students who used a z-value. A significant number performed a complicated calculation in part (b)(i), using the 2012 standard deviation and obtained the wrong answer. In part (b)(ii), a fair proportion of students recognised that the degree of overlap precluded any definite conclusion. Credit was given where students stated that there was some suggestion of a reduction in the mean, but others were often far too definite in their conclusions.

Q7.

Most candidates found all but part (b)(ii) of this question very challenging. In part (a), the common incorrect answers were verbose about university salaries or simply that 24.0 was too large when compared to 45.8. Even those candidates who calculated, for example $(45.8 - 2 \times 24.0)$, simply stated that a normal distribution, instead of salaries, could not take negative values. Answers to part (b)(i) were slightly better with candidates indicating a 'large sample' so 'Central Limit Theorem applied'. Some candidates lost both marks for 'if sample is large' whilst others lost a mark for suggesting that other than the sample mean could be assumed to be normally distributed. There were many correct answers to part (b)(ii) though some candidates lost marks for multiplying 24.0 by some combination of n and $(n - 1)$ or for an incorrect z-value. In answering part (c), a majority of candidates attempted the comparison of 55 with their CI or UCL. However, far too many then did the

same with 60 rather than compare $\frac{6}{50} = 12\%$ with 25%. This perhaps suggested that they had not read the question with sufficient care.

Q8.

Most candidates found much of this question very challenging. In part (a)(i), it was most disappointing to see the number of candidates failing to score 3 marks. Whilst there was some excuse for candidates dividing $\sum (x - \bar{x})^2$ by 50, rather than 49, there was really no

excuse for the many who did not know that $\bar{x} = \frac{\sum x}{n}$. Frequent incorrect attempts included trying to substitute $\sum x$ into $\sum (x - \bar{x})^2$ or dividing $\sum x$ by a variety of denominators. Some candidates were confused, either at the outset or later, by the units involved with invalid multiplications of s^2 by 1000 taking place at some point. In part (a)(ii), the common incorrect answers were verbose about university salaries or simply that 24.0 was too large when compared to 45.8. Even those candidates who calculated, for example $(45.8 - 2 \times 24.0)$, simply stated that a normal distribution or 'it', instead of salaries, could not take negative values. There were many partially correct answers to part (b) using follow-through answers from part (a)(i) though some candidates lost further marks for an incorrect Z-value. In answering part (c), a majority of candidates attempted the comparison of 55 with their CI or UCL. However, far too many then did the same with

60 rather than compare $\frac{6}{50} = 12\%$ with 25%, or considered the standardising of 60 using a normal distribution. This perhaps suggested that they had not read the question with sufficient care.

Q9.

The better students generally scored full marks in part (a) usually by working in litres. However, a large proportion of students provided answers that scored at most 2 marks. Errors seen were the use of 5 instead of 5.05, the use of 75 or 0.75 instead of 0.075, an incorrect z-value or the omission of the divisor of $\sqrt{36}$. The one mark for answering the

first claim in part (b) was only available to those better students who had obtained a correct confidence interval in part (a). Even so, some such students made no comparison using the stated mean of 5 litres but merely commented that the sample mean of 5.05 was greater than 5. This was very disappointing particularly given similar requests on previous papers. In answering the second claim in part (b), some students again referred to their

confidence interval and so scored no marks. Of those that correctly evaluated $\frac{8}{36}$ as 22%, many were not sufficiently explicit in comparing this with 10% and so scored only 1 of the 2 marks available.

Q10.

Many students still seemed not to understand whether to use t -values or z -values in constructing confidence intervals or carrying out hypothesis tests. In this question, the sample size was *small* and was selected from a *normal population with unknown variance* so t -values should have been used. Unfortunately many students neither used a t -statistic nor did they give their answers to one decimal place as requested. Consequently many more students than expected failed to gain full marks. In part (b), marks were lost by those students who failed to state that 40 did or did not, depending on the confidence interval constructed, lie in the confidence interval. It was not sufficient to simply state “Mean in CI” since this could have referred to 36.4 rather than to the value of 40.

Q11.

As usual, candidates found the calculations in parts (a) and (b) easier than the explanations in part (c), with the result that about 6 marks was the norm. Almost all candidates scored the 1 mark available in part (a)(i). However, more difficulties arose in answering part (a)(ii).

All too often, candidates evaluated $\sqrt{\frac{400.24}{65}}$ and then attempted to make it unbiased by

incorrectly multiplying by $\frac{65}{64}$ rather than $\sqrt{\frac{65}{64}}$. The best and simplest correct approach

was $\sqrt{\frac{400.24}{64}}$. Answers to part (b) showed a sound understanding and usually scored full marks.

Q12.

Surprisingly, this was probably the worst answered question on the paper, with even the better candidates scoring no more than about 9 of the 13 marks available. Most candidates scored the mark for answering part (a)(i) but many made a meal of the

calculation, rather than simply evaluating $\left(\frac{0.98 + 1.00}{2}\right)$.

Answers to part (a)(ii), which was essentially GCSE grade C material, were in the main very poor. All too often, the values quoted were 0.98 and 1.00, although numerous other incorrect values were seen. Even the better candidates, who managed to obtain the correct minimum value of 0.975, often quoted a maximum value of 1.004, 1.049, 1.0049 or even 1.05 rather than 1.005 or 1.0049.

Again in part (b), far too many incorrect answers were seen. A significant proportion of candidates used formulae, with varying success usually due to numerical or approximation errors. Even those candidates who used their calculators’ inbuilt statistical

functions obtained incorrect answers or quoted answers to insufficient accuracy. The most common incorrect answers were 1.065 and 0.073, obtained by using correct mid-points but ignoring frequencies.

Answers to part (c)(i) generally showed a sound understanding with at least 3 marks available despite incorrect answers in part (b). Incorrect z-values, omission of $\sqrt{n}$ or use of $\sqrt{8}$ were errors occasionally seen. Candidates exhibited a better understanding in answering part (c)(ii), perhaps because the idea of the use of mid-points for grouped data leading to an estimate of the mean was familiar from GCSE. Again, there was some confusion when candidates tried to discuss samples and populations, whilst some thought that the confidence interval was approximate because it was 99% rather than 100%.

Although correct answers, by comparing $\mu = 1$ with the confidence interval, were seen to part (d)(i), too many answers compared the sample mean with 1 litre or referred to 91 of the 100 cartons containing more than 1 litre. Again, in part (d)(ii), whilst some correct answers were in evidence, many candidates simply either noted that 1 litre was within the given interval or stated that 'it must contain all values as total probability is 1'. Some acceptable calculations of a normal probability were given, and a few candidates referred to $\bar{x} \pm 3s$ but rarely supported their statements with the necessary numerical justification.

Q13.

In part (a), it was expected that candidates would understand that for a sample of any size from a normal population with known variance, the reference distribution is $N(0, 1)$. Far too many candidates based their choice of distribution on the small value of n , and consequently incorrectly used t_{n-1} as their reference distribution. This resulted in a loss of all marks for this part of the question.

In part (b)(i), most candidates correctly used a t-distribution. Some candidates still used the z-distribution, with $z_{\text{crit}} = 6449$ often seen.

Although most candidates correctly found $\bar{x} = 550$, there were far too many who thought that $s\sqrt{\frac{334}{9}}$ or, even worse, $s = \sqrt{334}$.

In part (b)(ii), far too many candidates failed to state the level of significance, even though this was specifically asked for in the question. Others incorrectly thought that the level of significance was 90%. Many candidates did not justify their comments by correctly stating that 545 was not in the confidence interval; incorrect statements, such as 'it is not in the CI', gained no credit.

Q14.

It was of concern to see the large proportion of weak answers to this question, particularly as the topic has been examined in a similar way on all previous papers. The construction of the 96% confidence interval caused problems for far too many candidates. Common errors were an incorrect z-value (often 1.7507); use of 251 or 250 for $\bar{x}$; use of 1.92, $\sqrt{1.94}$, 1.94^2 or even 184.5 for s . Those few candidates who had obtained a correct answer in part (a)(i) then struggled in part (a)(ii) by failing to clearly compare 250 (the words 'it' or 'mean' were not acceptable) with their confidence interval in order to comment on the manufacturer's claim. Some stated incorrectly that their interval contained 250 whilst others stated that the claim was false since their interval did not contain 250. Answers to part (b)(i) rarely scored marks due to candidates failing to recognise that it required a repeat of part (a)(i) with $n = 1$. As a result, only the very best candidates were

able to gain the mark in part (b)(ii).

Q15.

It was most surprising and disappointing to see the large proportion of weak answers to this question, particularly as the topic has been examined in a similar way on all previous papers.

In part (a), more candidates started with $\frac{184.5}{49}$ rather than $\frac{184.5}{50}$ to obtain s^2 , but some then omitted to indicate the square root as the next step. Some candidates appeared to have no idea of the steps required, and either made no attempt or tried substituting $\bar{x} = 251.1$ into $\sum (x - \bar{x})^2 = 184$.

The construction of the 96% confidence interval caused problems for far too many candidates. Common errors were an incorrect z-value (often 1.7507); use of 251 or 250 for $\bar{x}$; use of 1.92, $\sqrt{1.94}$, 1.94^2 or even 184.5 for s ; divisors of 50 or even 1.

Some candidates substituted correct values into an incorrect formula, such as

$$z \pm \frac{s}{\sqrt{n}} \times \bar{x} \quad \text{or} \quad \bar{x} \pm \frac{zn}{s}$$

Most of these aforementioned errors lost most, if not all, of the 4 marks. Many candidates who had obtained a correct answer in part (b)(i) then struggled in part (b)(ii) by failing to clearly compare 250 (the words 'it' or 'mean' were not acceptable) with their confidence interval in order to comment on the manufacturer's claim. Some stated incorrectly that their interval contained 250 whilst others stated that the claim was false since their interval did not contain 250. Answers to part (c) rarely scored marks due to candidates' providing a qualitative judgement rather than a numerical justification. To score the 2 marks, candidates had to show that $[251.1 - (\geq 1) \times 1.94]$ gave a value less than 250 and so some bags were likely to be underweight.

Q16.

Overall this was the worst answered question on the paper with most answers revealing a marked lack of understanding of much of the material examined. As a result, very few candidates scored more than about 5 marks. Answers to part (a) were generally very disappointing, with most candidates at best only able to score 1 mark for finding $\bar{t}$ as an unbiased estimate of μ . The fact that the 50 times needed to be a random sample

$$\frac{\sum (t - \bar{t})^2}{49}$$

appeared unknown to most candidates, as did the fact that $\frac{\sum (t - \bar{t})^2}{49}$ was an unbiased estimate of σ^2 ; knowledge of the latter is a key area for future improvement. Failure to determine, or even attempt to determine, an unbiased estimate of σ^2 inevitably had a knock-on effect in part (b), as almost all candidates, whilst aware of the formula required, obtained an inaccurate or, more often, a totally incorrect confidence interval. In part (c), most candidates realised that they needed to compare 60 minutes with their intervals, but many then failed to translate their comparisons into a correct statement about Wyn's belief. Answers to part (d) revealed that an understanding of the correct interpretation of a confidence interval was not well understood. Common incorrect answers were 0, 1, or 0.99.

Q17.

Candidates were very well-prepared for part (a) and, as a result, most of them scored all 5 marks. Of the small minority who lost marks, the usual error was the use of an incorrect z -value. Here, the rounding of 1003.99 to 1003 or that of 1056.01 to 1057 were treated as irrelevant subsequent work and so were not penalised.

Many candidates scored the 2 marks in part (b) for a correctly reasoned comment on the claim. However, weaker candidates based their comments on a comparison of 1030 with their confidence intervals or suggested that the shovel always collected more than 1000 kg.

Q18.

Most candidates simply ignored the request for a necessary assumption in part (a) or stated that 'Heights' or 'Elephants', or 'The sample' or even 'They' were normally distributed. An answer needed to indicate that the 12 elephants were '*A random sample*' or were '*Selected independently*' or were '*Representative*'. Construction of the confidence interval was invariably correct. However a small minority of candidates lost marks for using an incorrect formula, a calculated value of s^2 rather than 0.20 or even an incorrect value for the sample mean.

In part (b), many candidates lost marks because their answers did not justify their (often too definitive) statements. A clear comparison of 2.90 with a calculated confidence interval was required and conclusions had to refer to mean heights, not heights in general.

Q19.

In part (a), candidates were asked to state any assumption that they had made in constructing their confidence interval. Candidates should be aware that a meaningful interval cannot be constructed in this case unless they assume that they are dealing with a small sample from a normal population with unknown variance. It should also be noted that the assumption should be made with the context of the question in mind. Simply stating "Normal distribution" or "It is normal" or "the population is normal", without also stating what this population is, will not gain full credit. Also, many candidates incorrectly thought that the sample or the data had to be normally distributed.

Those candidates who did not state an assumption still calculated a confidence interval based upon the t -distribution. Most candidates calculated correctly that $\bar{v} = 117.9$ but were often less sure when it came to calculating an unbiased estimate, s^2 , for the variance

$$s^2 = \frac{1014.9}{9} = 112.8$$

of the population. Although a majority calculated correctly that (or $s = 10.6$), there were many who thought wrongly that $s^2 = 101.49$ (or $s = 10.1$) or $s =$

$\sqrt{1014.9} = 31.9$. Those who stated that $\sigma = 10.1$, nearly always went on to wrongly state

that $\frac{10.1}{\sqrt{10}}$ was the value for the standard error. Appropriate use of the formulae booklet provided could have eliminated many of these errors.

The majority of candidates who calculated the correct values for $\bar{v}$ and s then used $t = 3.250$ and went on to construct the required confidence interval as (107, 129).

Occasionally, the wrong value of t was extracted from the tables, and there were some candidates who thought that the reference distribution was $N(0, 1)$, rather than t_{n-1} ; this lead them to use wrongly a variety of z -values in their calculations.

It seemed, from the responses to part (b), that very few candidates understand what a confidence interval shows. Many candidates firstly stated that John's claim was "wrong"

and then indicated that this was because 99% of his serves were between 107 mph and 129 mph. Candidates should be aware that, when studying this section of the specification, emphasis should be given to both the construction and the interpretation of confidence intervals.

Q20.

This question proved to be the most difficult on the paper. In part (a), many candidates made a complete hash of the deduction process by apparently not realising that 1 hour was equivalent to 60 minutes. Thus it was all too common to see candidates adding 1, or even 100, to both mean and standard deviation values or multiplying one or both values by 60 or 100. Future candidates clearly need to be much better prepared for similar questions involving linear scaling. In answering part (b), most candidates were clearly aware of the relevant formula for a confidence interval and identified that $z = 2.3263$ but lost at least 1 accuracy mark due to the aforementioned errors in part (a) or by simply using the given values of 1.90 and 3.32. The awarding of the mark in part (c) was rare indeed since the answer required a comparison of 1 hour or 60 minutes with a correspondingly **correct** confidence interval in part (b).

Q21.

This question proved to be the most difficult on the paper. It was very disappointing to see so many candidates, some scoring high marks on other questions, failing, either by using their calculators' statistical functions or from first principles using formulae, to find correct values in part (a)(i). All too often, incorrect mid-points were identified or, more seriously, frequencies were ignored. Attempts at part (a)(ii) were often equally disappointing. Whilst some candidates ignored the word 'Hence' and so repeated part (a)(i) with revised mid-points, many of those who tried to deduce answers made a complete hash of the process by apparently not realising that 1 hour was equivalent to 60 minutes. Thus it was all too common to see candidates adding 1, or even 100, to both mean and standard deviation values or multiplying one or both values by 60 or 100. Future candidates clearly need to be much better prepared in these areas of the specification. In answering part (b)(i), most candidates were aware of the relevant formula for a confidence interval but lost 1 accuracy mark due to the aforementioned errors in part (a). Candidates who used an incorrect z -value or, more seriously, used $n = 8$, lost more marks here. In part (b)(ii), the most common acceptable reasons were "actual times unknown" or "midpoints used" whereas the common response of "mean and standard deviation are estimates" did not gain the mark available. The scoring of marks in part (c) was rare indeed. Most candidates failed to identify the difference between Vernon's two claims and so referred to their sample mean values for both. His first claim required reference to 74% (or 26%) from the original data, whereas his second claim required a comparison of 1 hour or 60 minutes with a correspondingly **correct** confidence interval in part (b)(i). A small minority of candidates considered that the claims were best answered by commenting on Vernon's plumbing skills and work ethic.

Q22.

This was, somewhat surprisingly, the worst answered question on the paper. A **small** sample from a normal distribution with **unknown** variance required use of the t -distribution. Consequently those candidates, of whom there were far too many, who used $z = 2.5758$ instead of $t = 3.250$ inevitably lost marks here. The better candidates

found $\bar{x} = 3.19$ and then correctly used $s^2 = \frac{1.849}{9} = 0.2054$ in constructing the 99%

confidence interval by evaluating $3.19 \pm t_9 \times \frac{\sqrt{0.2054}}{\sqrt{10}}$ to give (2.72, 3.66). Unfortunately,

some candidates, who thought incorrectly that $s^2 = \frac{1.849}{10} = 0.1849$ and so used $3.19 \pm t^9$
 $\times \frac{\sqrt{0.1849}}{\sqrt{10}}$ instead of $3.19 \pm t^9 \times \frac{\sqrt{0.1849}}{\sqrt{9}}$, failed to find the correct confidence interval.
 In part (a)(ii), a statement that '3.5 is within the confidence interval' with a conclusion that 'the claim was therefore reasonable' was all that was required here. Part (b) was usually done well, except by those candidates who thought that 1% was equivalent to finding $\frac{1}{10} \times 200 = 20$.

Q23.

Most candidates scored at least 4 marks. The formula for the confidence interval was well-known and used correctly by most candidates; an incorrect z-value was the usual error. In answering part (b)(i), many comments observed that the mean for this play lay inside the confidence interval (as of course it must) and so the assumption was valid. This suggested anticipation on the part of these candidates of the type of comment that has been expected following the determination of a confidence interval on previous papers. In part (b)(ii), plays having different popularities or audience sizes were the most frequently seen acceptable reasons for the comment "Unlikely to be valid".

Q24.

It was most gratifying to note the many candidates scoring 14 or more marks (out of 16) on this question. In part (a)(i), 'corrections to 40' were thankfully very rare with the result that most candidates scored full marks. Again in part (a)(ii), full marks was the norm although a small minority of candidates failed to allow for the negative sign of 1.6 and so evaluated $0.94520 - 0.65542$ (loss of all 3 marks). The most common mistake in part

(a)(iii), seen even on the best scripts, was to equate $\frac{x - 38}{5}$ to $+ 0.6745$ instead of -0.6745 (loss of 1 mark).

Despite comments on previous Examiner Reports, there still remain candidates who are apparently unaware of Table 4.

There were many correct answers given for the confidence interval in part (b)(i). The most common error was the use of an incorrect z-value, often 2.0537 (Table 4 yet again).

Thankfully very few candidates omitted to divide the standard deviation by $\sqrt{40}$. In part (b)(i), candidates had to first identify that $2.5 \times 38 = 95$ before any marks were awarded.

Most candidates did indeed do this but a few evaluated $\frac{107}{2.5}$. Having obtained the value of 95 candidates then had to compare it against their confidence interval, not 107, in order to gain marks for a (follow-through) correct conclusion. Thus it was pleasing to see the many instances of candidates scoring all 3 marks.

Q25.

A large majority of candidates scored full marks on this question. In part (a), a minority of candidates used an incorrect z-value, usually 1.6449, or, more critically, omitted to divide by $\sqrt{50}$. An 'adjustment' to the value of 25.1 for the sample standard deviation was not expected and was rarely, if ever, seen. Similarly, use of the t-distribution (Statistics 2) was very rare but not penalised when applied correctly. Almost all candidates stated a valid

reason in part (b), usually referring to 'selection by size of potato'. Unacceptable reasons usually referred to a small sample size.

Q26.

The numerical answers to part (a) were usually correct. However, there were many candidates who did not give an assumption or did not know what assumption they had made. The assumption that "The sample is normally distributed" was the usual false statement and "It is normally distributed" or simply "Normally distributed" were also insufficient. In fact it was the population from which the sample was taken that had to be assumed to have a normal distribution.

In part (b), a popular misconception by candidates seemed to be that the 99% confidence interval (29.2, 42.0) indicated that 99% of motorists travelled at speeds of between 29.2 mph and 42.0 mph, whereas it really indicated that there was a 99% confidence that the interval contained the true mean speed of all motorists travelling through the village. Whilst the majority of candidates indicated that 30 mph fell within the confidence interval, most then failed to realise that 80% of the sample had speeds in excess of 30 mph which consequently indicated that the speed limit was not being adhered to by most motorists.

Q27.

Statistics 1A

As mentioned earlier, far too many candidates lost at least 1 mark in part (a). Those that did not, usually used s_n rather than s_{n-1} whereas those that did, used s^2n , s^2n-1 ,

$$\sqrt{\sum (x - \bar{x})^2} \text{ or even } \sum (x - \bar{x})^2.$$

In part (b), the value of z was identified correctly but the divisor of $\sqrt{n}$ was omitted on too many occasions; perhaps candidates were wary after already dividing by n in part (a)? Those candidates who used 286.5 for their sample mean apparently saw nothing wrong with a clearly incorrect answer if the context was considered! The many candidates who attempted parts (a) and (b) correctly invariably provided correct answers to an appropriate level of accuracy. Those candidates who referenced their confidence interval in answering part (c) usually scored the 2 marks available. However, too many candidates merely compared the claims with their value for the sample mean or produced a qualitative argument based upon their perceived reasons as to why the fire and rescue service would underestimate whilst the parish councillor would overestimate.

Statistics 1B

As mentioned earlier, far too many candidates lost at least 1 mark in part (a). Those that did not, usually used s_n rather than s_{n-1} whereas those that did, used s^2n , s^2n-1 ,

$$\sqrt{\sum (x - \bar{x})^2} \text{ or even } \sum (x - \bar{x})^2.$$

In part (b), the value of z was identified correctly but the divisor of $\sqrt{n}$ was omitted on too many occasions; perhaps candidates were wary after already dividing by n in part (a)? Those candidates who used 286.5 for their sample mean apparently saw nothing wrong with a clearly incorrect answer if the context was considered! The many candidates who attempted parts (a) and (b) correctly invariably provided correct answers to an appropriate level of accuracy. The many candidates who referenced their confidence interval in answering part (c) usually scored the 2 marks available. However, too many merely compared the claims with their value for the sample mean.

Q28.

Most candidates were able to first show that $\bar{x} = 301.0$ and $s^2 = 24.5$. In part (b), although there were some good solutions seen, too many candidates used a z -value rather than the required t -value. Responses to part (c) were rather disappointing with comments often not in context and usually with no reference to the confidence interval that they had just found in part (b).

Q29.

There were many excellent solutions seen to part (a), with most candidates realising that they first had to find that $\bar{x} = 8.0$ and that $s = 2.121$. Although many candidates understood that the t -distribution had to be used with $\nu = 8$ and hence $t = 1.860$, since the data given formed a small sample from a normal distribution with unknown standard deviation, many others did not. These latter candidates usually used z -values from the normal distribution tables. Part (b) was not well answered as many candidates failed to make any reference to the context of the question or the confidence interval that they had just found.

Q30.

Many candidates scored full marks in part (a) and it was pleasing to observe the improvements in notation and presentation. On the rare occasions when marks were lost in part (a)(i), it was for standardising 1014 or 1014.5, rather than 1015; something that needs to be eradicated for the future. In part (a)(ii), some candidates obtained the correct final answer but not always by the most direct method; many others usually failed to carry out the necessary area change. In part (a)(iii), far too many candidates assumed that all that was required was to find the difference of their two previous answers and so scored no marks. Candidates who missed the link and so began afresh, often scored full marks. Candidates need to think more carefully about such linked parts to questions.

In part (b)(i), few candidates scored all 6 marks. Despite comments in a previous Examiners Report and the relevant formula for s^2 on page 10 of the supplied booklet, far too many candidates were unable to calculate the correct value for s from the given value for $\sum (y - \bar{y})^2$; something that must be addressed for the future. Other less frequent errors were due to an incorrect value for $\bar{y}$, an incorrect z -value or omission of $\sqrt{50}$. In part (b)(ii), the awarding of full marks was even rarer. Many candidates did not make it clear that they were comparing 500 (and not the sample mean) with their confidence interval, failed to indicate that 6 in 50 packets were underweight or stated no overall conclusion.

Q31.

In part (a)(i), many candidates scored all 5 marks. When this was not the case, it was usually due to an incorrect z -value or the omission of $\sqrt{10}$. In part (a)(ii), the awarding of full marks was rare. Many candidates did not make it clear that they were comparing 500 (and not the sample mean) with their confidence interval, failed to indicate that 3 in 10 packets were underweight or stated no overall conclusion. Correct answers to part (b) were also rare. The word 'Hence' together with an allocation of 1 mark should have suggested that lengthy working, as produced by many candidates, was not required.

Q32.

Many candidates appeared ill-prepared for this type of routine question. As a result, in part (a) the success rate on the straightforward confidence interval calculation was disappointing. Most candidates found the correct value for the sample mean but some candidates used a z-value of 2.0537, rather than 2.3263, whilst others omitted the divisor $\sqrt{12}$ or used the sample standard deviation instead of the given value of 3.5. However, in general, well-prepared candidates scored all 5 marks.

In answering part (b), most candidates paid no heed to the fact that 3 marks were available and so opted for general subjective assumptions that usually scored no marks. Thus few candidates made use of their calculated confidence interval and even less gave a correct interpretation of it. There was also minimal reference to the sample data. As a result, even the best candidates, who recognised the relevance of both the sample data and their confidence interval, were reluctant to commit to a conclusion.

Q33.

Most candidates were able to answer part (a) competently, often immediately spotting that the answer to part (ii) involved the answer to part (i). Weaker candidates often wanted to insert a continuity correction and a small minority had confused ideas about variance and standard deviation. In part (b)(i), confusion again arose over variance and standard deviation. Few candidates divided 533.61 by 49 immediately. Some candidates used a

correction factor of $\frac{n}{n-1}$ later, but many forgot or applied it to the standard deviation. A fair sized minority of candidates did not divide by $\sqrt{50}$ when calculating their confidence interval for the mean. Most candidates correctly used z in the acceptable range of 2.57 to 2.58. Overall, few candidates scored the full six marks on this part with most scoring in the range three to five marks. A few candidates were apparently unconcerned when they found that Mustafa took upwards of 14 hours to walk to school! Only the better candidates could make a sensible attempt at answering part (b)(ii). Many candidates did not appreciate the need to add 25% onto 12 and so obtain 15, which then had to be looked at in the light of their confidence interval. Those candidates who did so, were then usually able to score all three marks.

Q34.

Part (a) proved a good source of marks with a pleasing proportion of completely correct solutions. A small minority of candidates used a z-statistic instead of the appropriate t-statistic. Most candidates were able to deduce the required confidence interval in part (b)(i) and to use it in support of a sensible comment in part (b)(ii).

Q35.

As mentioned previously, answers to part (a) were generally very disappointing. Most candidates making any progress divided by 50 though about a quarter subsequently

scaled their answer using $\frac{50}{49}$. Candidates should be encouraged to find unbiased estimates of σ^2 using $\frac{s^2}{n-1}$ directly. Many candidates did use a correct expression for standard error but many omitted the square root in either the numerator or denominator.

Follow-through marks in part (b)(i) enabled many candidates to score at least four marks but the weakest candidates failed to evaluate $\bar{x}$ or find a correct z-value.

Answers to part (b)(ii) generally scored two marks for a correct deduction based on an

answer to part (b)(i) though again clarity left much to be desired.

In part (c), whilst many candidates realised that the value of μ had increased/changed, only a handful of candidates referred to non-overlapping intervals for the second mark. Often comments suggested the new confidence interval was more reliable due to an increased sample size.

Q36.

This question elicited good answers from most candidates and there were few common errors. In part (a), some candidates used $z = 3$ whilst others calculated an unbiased estimate of variance from the twelve values or tried to adjust the given population standard deviation for bias. Most candidates who found a confidence interval in part (a) were then able to make a sensible (follow-through) attempt at part (b), though a small minority compared 410 with 413 or with the twelve individual values.

Q37.

This question was well answered by the majority of candidates. Almost without exception, all but the weakest candidates scored the 3 marks in part (a)(i) and there were few attempts at a continuity correction. In part (a)(ii), whilst method marks were often scored, candidates quoted an inaccurate z -value possibly due to use of Table 3 rather than Table 4 and/or gave an incorrect sign. Some candidates used a 'non z -value' of $1 - 0.67$ (0.1) or simply 0.75. In parts (b)(i) and (ii), some candidates biased their estimate of standard

error, some even of their mean, by multiplying by $\left(\frac{n}{n-1}\right)$. However, these candidates and most others then scored full marks in part (b)(iii) by ignoring such attempted corrections, though a small minority used either 2 or 1.645 rather than 1.96. Answers to part (b)(iv) were usually sufficiently clear with correct deductions from previous answers.

However, some candidates suggested that 24.94 would round to 25 or that 25 was close to the sample mean.

Q38.

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All candidates determined correctly μ as $\bar{x}$. However some candidates found a biased estimate of σ^2 ($\div n$) whilst others took the long route of calculating the sample variance and

then multiplying it by $\left(\frac{n}{n-1}\right)$. Calculations of the confidence interval in part (a)(i) were very sound, with the use of z , rather than t , the only error of note. All candidates knew what was required in part (b) and scored the 2 marks available.

Q39.

Apart from candidates who thought that the mean amount spent by each of the previous 400 customers was £4256, there were few problems with parts (a) and (d)(i).

Part (b) was almost universally well answered apart from a few candidates who confused standard deviation and variance.

There were many reasonable attempts at part (d)(i) but very few candidates made the connection between the null hypothesis $\mu = 11$ in part (d)(i) and the upper limit of the confidence interval in part (a). A variety of sensible suggestions were made in part (d)(iii)

Q40.

Apart from candidates who thought that the mean amount spent by each of the previous 400 customers was £4256, there were few problems with parts (a) and (b)(i).

In part (b)(ii) however, many candidates demonstrated that they completely misunderstood the meaning of a confidence interval. Often they did not realise that it has any connection with the mean but thought that 95% of the amounts spent on petrol would lie within the limits of the confidence interval. Many made sensible suggestions in part (b)(iii)

Q41.

Parts (a) and (b) were well answered. Unlike previous years, there was a mark in part (a) for giving the answer to an appropriate degree of accuracy (4, 5 or 6 significant figures were accepted). Those candidates who hedged their bets by giving alternative answers to different levels of accuracy did not receive this mark unless all their answers were to 4, 5 or 6 significant figures.

Most candidates made reasonable attempts at parts (c) and (d), although answers to part (d) revealed that many candidates did not realise that confidence intervals apply to means and are not intervals within which a given percentage of the population lies.

Q42.

Parts (a) and (b) were well answered. Unlike previous years, there was a mark in part (a) for giving the answer to an appropriate degree of accuracy (4, 5 or 6 significant figures were accepted in this case). Those candidates who hedged their bets by giving alternative answers to different levels of accuracy did not receive this mark unless all their answers were to 4, 5 or 6 significant figures.

Most candidates made reasonable attempts at parts (c) and (d), although answers to part (d) revealed that many candidates did not realise that confidence intervals apply to means and are not intervals within which a given percentage of the population lies.

Q43.

Parts (a) and (b)(i) were well answered. The great majority of candidates successfully obtained standard deviation directly from their calculators as recommended. In part (a), a few candidates failed to use t , or used the wrong divisor for the standard deviation.

Only a minority of candidates were able to answer part (b)(ii) correctly, many merely repeated their answer to part (b)(i). Others calculated the width of the confidence interval.

The interpretation in part (c) was generally poor. Many candidates failed to make any connection between the confidence interval and the mean, believing rather that 95% of scores lie in the 95% confidence interval. The answers to parts (d) and (e) were better. In part (e), there was some concern as to the validity of combining two samples. However, as both samples were stated to be random samples the combined sample would also be a random sample.

Q44.

Parts (a) and (b)(i) were well answered. Only a minority of candidates were able to answer part (b)(ii) correctly, many merely repeated their answer to part (b)(i). Others, apparently influenced by a question on a previous paper, calculated the width of the confidence interval.

The interpretation in part (c) was generally poor. Many candidates failed to make any connection between the confidence interval and the mean, believing rather that 95% of scores lie in the 95% confidence interval. The answers to parts (d) and (e) were better. In part (e) there was some concern as to the validity of combining two samples. However as both samples were stated to be random samples the combined sample would also be a random sample.

Q45.

Most candidates were well prepared for part (a) although a few failed to use the t-distribution. Part (b) was not generally well answered, particularly part (b)(ii) where only a small minority pointed out that it is not possible to consume negative amounts of water and so the normal distribution cannot provide an exact mode.

All parts of (c) were well answered. It was expected that, as the sample was large, candidates would use z-values and the given standard deviation. Some used t-values.

Some interpreted the standard deviation as $\sqrt{\frac{\sum (x - \bar{x})^2}{n}}$ and applied a correction to obtain their value.

Most candidates made good attempts at parts (c)(i) and (c)(ii) although quite a few stated that because the sample was large the distribution (rather than the distribution of the sample mean) would be approximately normal.